

BST A1000L

Data sheet

V 3.1

Black Sesame Technologies Co., Ltd.

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Index

1 Introduction	1
1.1 Block diagram	2
1.2 Ordering information	2
2 Abbreviated Terms	4
3 Features Summary	5
3.1 CPU cluster	5
3.2 GPU	5
3.3 Memory and Storage	5
3.4 Image sensor Interface	5
3.5 ISP	5
3.6 Video codec	6
3.7 VOUT Interface	6
3.8 Computer Vision(CV)	6
3.9 Neural Network(BSNN)	7
3.10 Safety and Security	7
3.11 High-speed IO Interfaces	8
3.12 DMA	8
3.13 Peripheral Interfaces	8
3.14 Package	9
3.15 ISO26262 Information	9
4 Electrical characteristics	10
4.1 Chip-level conditions	10
4.1.1 Absolute maximum ratings	10
4.1.2 Thermal resistance	11
4.1.3 Operating ranges	11
4.1.4 External clock sources	12
4.1.5 Maximum supply currents	13
4.1.6 Power modes	14
4.1.7 USB PHY DC electrical characteristics	14
4.1.8 PCIe PHY DC electrical characteristics	14
4.1.9 MIPI DPHY DC electrical characteristics	15
4.1.10 DDR	16
4.2 Power supplies requirements and restrictions	18
4.2.1 Power-up sequence	18
4.2.2 Power-down sequence	19
4.2.3 AC noise requirement	19
4.3 PLL electrical characteristics	20
4.4 I/O DC parameters	21
4.4.1 General purpose I/O (GPIO) DC parameters	21
4.4.2 SDIO DC parameters	23

4.4.3 DDR I/O DC electrical characteristics	25
4.5 I/O AC parameters	27
4.5.1 General purpose I/O AC parameters	28
4.5.2 SDIO AC parameters	28
4.5.3 DDR I/O AC parameters	29
4.6 System modules timing	32
4.6.1 Reset timings parameters	32
4.6.2 WDOG Reset timing parameters	32
4.7 External peripheral interface parameters	32
4.7.1 JTAG timing parameters	32
4.7.2 UART timing parameters	33
4.7.3 I2C timing parameters	34
4.7.4 SPI timing parameters	35
4.7.5 I2S timing parameters	35
4.7.6 PWM timing parameters	37
4.7.7 OSPI timing parameters	37
4.7.8 CAN FD timing parameters	39
4.7.9 SDEMMC timing parameters	39
4.7.10 ERR_RPT timing parameters	41
4.7.11 GMAC timing parameters	41
4.7.12 VIN timing parameters	46
4.7.13 VOUT timing parameters	47
4.7.14 USB3 PHY parameters	48
4.7.15 USB2 PHY parameters	49
4.7.16 PCIE PHY parameters	49
4.7.17 MIPI DPHY parameters	51
5 Boot configuration	53
5.1 Boot mode configuration pins	53
5.2 Boot device interface allocation	53
5.3 Boot strap pin configuration	53
6 PCB Guideline or Requirement	55
6.1 PCB Guideline	55
6.2 Unused pin	55
7 Package information and contact assignments	56
7.1 Package outline Dimension	56
7.2 Supplies contact assignments and functional contact assignments	57
7.3 Pinmux and Ball map	58

Tables

Table 1-1	Ordering information	3
Table 2-1	Abbreviated Terms	4
Table 4-1	Absolute maximum ratings	10
Table 4-2	Thermal Resistance Characteristics	11
Table 4-3	Operating ranges	11
Table 4-4	Maximum supply currents	13
Table 4-5	Power domain	14
Table 4-6	Worst-Case Maximum Current	14
Table 4-7	Maximum Power Supply Slew Rates	14
Table 4-8	Supported Power Supply Limits at the PHY Macro Bumps	15
Table 4-9	Input DC Specifications (Apply to CLKP/N and DATAP/N Inputs)	15
Table 4-10	HS Line Receiver DC Specifications	15
Table 4-11	LP Line Receiver DC Specifications	16
Table 4-12	Recommended Operating Conditions ¹	16
Table 4-13	A Power-up Sequence Example	18
Table 4-14	AC noise	19
Table 4-15	PLL electrical characteristics	20
Table 4-16	Recommended Operating Conditions	21
Table 4-17	DC Characteristics	21
Table 4-18	Recommended Operating Conditions	23
Table 4-19	DC Characteristics	23
Table 4-20	Common DC Input Conditions	25
Table 4-21	Common DC Input Conditions for Flyover Bypass	25
Table 4-22	LPDDR4 DC Input Conditions	25
Table 4-23	Common DC Output Parameters	26
Table 4-24	Maximum frequency of operation	28
Table 4-25	Maximum frequency of operation	29
Table 4-26	Maximum frequency of operation	29
Table 4-27	Common AC Input Conditions	29
Table 4-28	LPDDR4 AC Input Conditions	29
Table 4-30	Common AC Output Parameters	30
Table 4-31	Output Timing Characteristics	31
Table 4-32	Common DC Output Conditions for Flyover Bypass ¹	31
Table 4-33	Common DC and AC Output Conditions for Reset_n ¹	31
Table 4-34	Table Reset timing parameter	32
Table 4-35	Table Reset timing parameter	32
Table 4-36	Timing Requirements for IEEE 1149.1 JTAG	33
Table 4-37	Switching Characteristics Over Recommended Operating Conditions for IEEE 1149.1 JTAG	33
Table 4-38	Serial mode transmit timing parameters	34
Table 4-39	RS-232 Serial mode receive timing parameters	34
Table 4-40	SPI master timing parameter	35

Table 4-41	Master transmitter with data rate of 6.144MHz ($\pm 10\%$) (all values in ns)	36
Table 4-42	Slave receive with data rate of 6.144MHz ($\pm 10\%$) (all values in ns)	36
Table 4-43	Timing parameter for i2s transmitter and receives	37
Table 4-44	PWM output timing parameters	37
Table 4-45	OSPI timing parameter	38
Table 4-46	DS/HS/SDR12/SDR25/SDR50 mode timing parameter	40
Table 4-47	EMMC bus speed modes summarizes	40
Table 4-48	Timing parameters of EMMC	40
Table 4-49	GMAC signals map	41
Table 4-50	RGMII signal switching specifications	43
Table 4-51	GMII signal switching specifications	44
Table 4-52	MII signal switching specifications	44
Table 4-53	MDIO signal switching specifications	45
Table 4-54	VIN signals map	46
Table 4-55	VIN timing parameters	46
Table 4-56	VOOUT signals map	47
Table 4-57	VOOUT timing parameters	47
Table 4-58	Reference Clock Requirements	48
Table 4-59	Reference Clock Jitter Requirements for PCIe 1.1 and 2.1	49
Table 4-60	Reference Clock Jitter Requirements for PCIe 3.1	49
Table 4-61	Reference Clock Electrical Characteristics	49
Table 4-62	Reference Clock External Pad Input Parasitics	50
Table 4-63	Reference Resistor Requirements	50
Table 4-64	AC Coupling Capacitors - Consumer 10G AP PCIe 3.0 PHY Protocol	51
Table 4-65	Example Supply Bypass Capacitors	51
Table 4-66	External Component Requirements	51
Table 4-67	High Speed Clock Timings	51
Table 5-1	Boot Mode Configuration	53
Table 5-2	Interface allocation during boot	53
Table 5-3	Boot strap pin configuration	53
Table 5-4	Strap PINs for PCIe	54
Table 7-1	A1000L supplies contact assignments	57

Figures

Figure 1-1	Block Diagram	2
Figure 1-2	Marking Diagram	2
Figure 4-1	Core Voltage Tolerance for Minimum VDD Voltage	17
Figure 4-2	Core Voltage Tolerance for Maximum VDD Voltage	17
Figure 4-3	Load circuit for output	28
Figure 4-4	Output transition time waveform	28
Figure 4-5	Reset timing diagram	32
Figure 4-6	WDOG timing diagram	32
Figure 4-7	JTAG Timing	33
Figure 4-8	Serial data format	34
Figure 4-9	UART RS-232 Serial mode transmit timing	34
Figure 4-10	UART RS-232 Serial mode receive timing	34
Figure 4-11	SPI input timing	35
Figure 4-12	SPI output timing	35
Figure 4-13	I2S bus stereo frame format	35
Figure 4-14	Timing for I2S transmitter	36
Figure 4-15	Timing for I2S receiver	36
Figure 4-16	PWM timing	37
Figure 4-17	SDR mode input timing	38
Figure 4-18	SDR mode output timing	38
Figure 4-19	CANFD bus sample timing	39
Figure 4-20	SD input timing	39
Figure 4-21	SD output timing	39
Figure 4-22	Data input/output timing	40
Figure 4-23	RGMII transmit signal timing diagram	43
Figure 4-24	RGMII receive signal timing diagram	43
Figure 4-25	GMII transmit signal timing diagram	44
Figure 4-26	GMII receive signal timing diagram	44
Figure 4-27	MII transmit signal timing diagram	45
Figure 4-28	MII receive signal timing diagram	45
Figure 4-29	MDIO transmit signal timing diagram	45
Figure 4-30	MDIO receive signal timing diagram	45
Figure 4-31	VIN timing diagram	46
Figure 4-32	VOUT timing diagram	47
Figure 4-33	Reference Clock External Pad Input Parasitics	50
Figure 7-1	Package outline dimension	57

1 Introduction

BST A1000L is a high-performance, low-power, real-time computer vision and neural network processing Automotive SoC, designed for a wide range of embedded applications with a central focus on camera-based sensing and AI computing for Advanced Driver Assistance System (ADAS) and Autonomous Driving Systems, based on TSMC 16nm process. BST A1000L SoC deploys a multi-core compute architecture consists of:

- Application processor cluster based on multi-core ARM Cortex-A family processor IPs
- High effective, low power, programmable Neural Network Acceleration Engine (NPU)
- The Image Signal Processing (ISP) pipeline IP core, which designed for digital processing and image quality enhancements of an input video stream
- Multi-threaded RISC-V real-time control unit (RCU) with specialized instruction set (ISA) extensions, which is used to control whole ISP flow
- Wide SIMD Vision DSP for improving flexibility of CV/AI processing

In order to reduce power consumption and improve computation efficiency, BST A1000L SoC deploys a highly optimized memory hierarchy that includes several smart memory modules and a multi-port DDR memory sub-system. The smart memory modules wrap around large on-chip SRAMs with intelligent DMA and memory management engines to improve on-chip memory utilization and minimize memory access traffic to DDR, therefore keeping the computation engines highly utilized at very low power consumption. In order to ensure the overall SoC performance and platform scalability, BST A1000L SoC includes dual 32-bit LPDDR4 high performance memory interfaces. It integrates multiple high-speed and peripheral interfaces, which can be tailored and configured to fit into different system level usage models.

To meet automotive function safety and security requirements, BST A1000L SoC deploys a dedicated safety and security sub-system that meets ASIL-D requirements. Several other modules involved in sensing for autonomous driving are designed according to ASIL-B requirements.

1.1 Block diagram

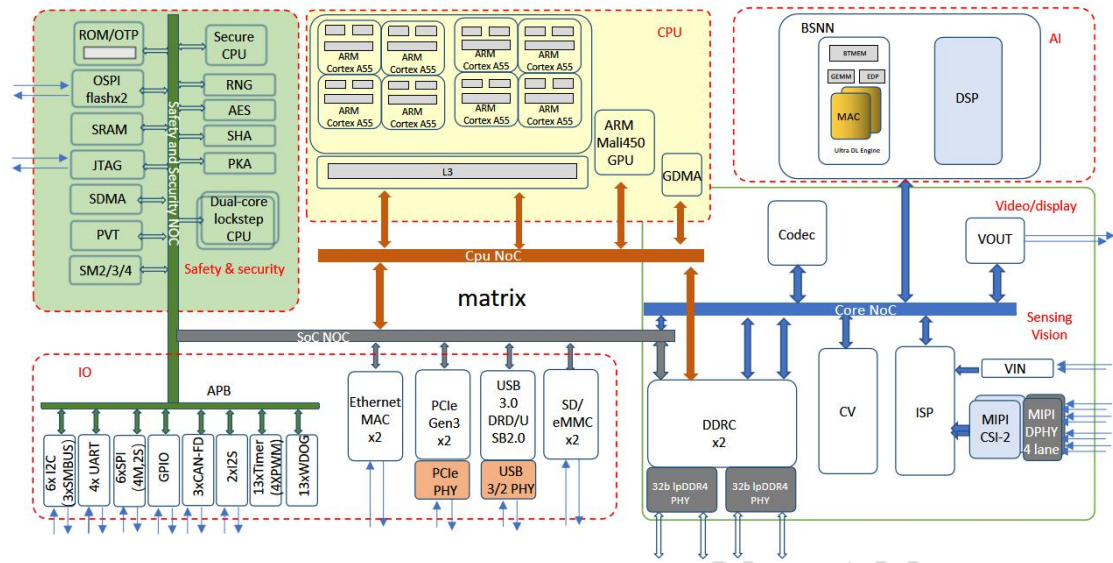


Figure 1-1 Block Diagram

1.2 Ordering information

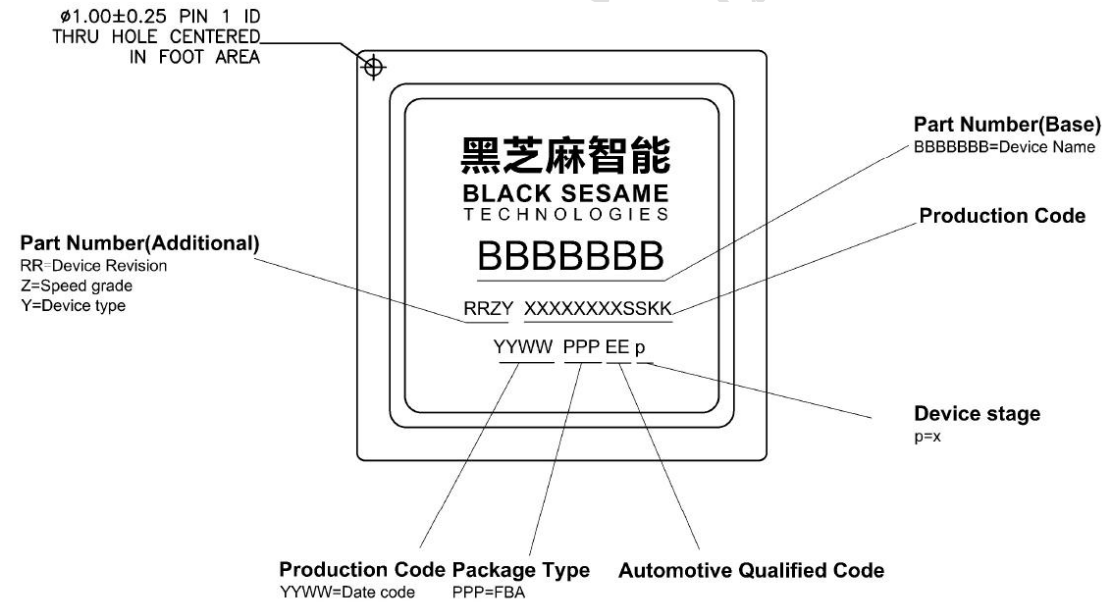


Figure 1-2 Marking Diagram

Table 1-1 Ordering information

Field Parameter	Field Description	Values	Value Description
BBBBBBBB	Base part number	A1000L	
RR	Device revision	As Marked	
Z	Speed grade	X	Default, 800MHz
		As Marked	Others
Y	Device type	X	Default
		As Marked	Others
XXXXXXXXSSKK	Production code 1	As Marked	
YY	Production code 2	As Marked	Year Code
WW	Production code 3	As Marked	Week Code
PPP	Package Type	FBA	FCBGA
EE	Automotive qualified	Q1	AEC-Q100 grade 2 qualified
		BLANK	Not automotive qualified
p	Device stage	x	Default

2 Abbreviated Terms

Table 2-1 Abbreviated Terms

Abbreviation	Description
ASIL	Automotive Safety Integrity Level
BIST	Built In Self Test
CAN	Controller Area Network
CAN FD	CAN with Flexible Data-Rate
CPU	Central Processing Unit
CRC	Cyclic Redundancy Check
CV	Computer Vision
DDR	Double Data Rate
DMA	Direct Memory Access
DSP	Digital Signal Processing
GPIO	General Purpose I/O
GPU	Graphics Processing Unit
I2C	Inter-Integrated Circuit
I2S	Integrated Interchip Sound
MIPI	Mobile Industry Processor Interface
OTP	One Time Password
PCIe	PCI Express
PLL	Phase Lock Loop
RAM	Random Access Memory
ROM	Read Only Memory
SOC	System-on-chip
SPI	Serial Peripheral Interface
UART	Universal Asynchronous Receiver Transmitter
PVT	Process, voltage and temperature sensor

3 Features Summary

3.1 CPU cluster

- Octa-core ARM Cortex-A55 processor @1.2GHz maximum frequency
 - 32KB I- and D- L1 caches in each core
 - 128KB combined L2 cache per core
 - 2MB shared L3 cache
 - NEON SIMD, FPU and crypto support

3.2 GPU

The ARM Mali-450 is optimized for applications that require UI or 3D graphics rendering support. The capability of GPU is sufficient for common AVM application and 1080p resolution UI.

- Mali-450 MP2 @1Ghz maximum frequency
- Up to 218 Mtri/s Triangle-Fillrate
- Up to 2 Gpix/s Pixel-Fillrate
- Up to 30 Gflops

3.3 Memory and Storage

- Dual 32-bit LPDDR4 memory interfaces @ 3200MHz maximum frequency, Support Inline ECC protection
- 2 OSPI Flash @100Mhz, up to 4GB, support datawidth 1x, 2x, 4x, 8x
- 2x SD/eMMC external storage interface(s)
- ~18MB on-chip SRAM distributed in CPU caches and smart memory blocks BSMEM and BTMEM
 - 8MB BTMEM
 - 1.5MB BSMEM
- 32K boot ROM

3.4 Image sensor Interface

- Two 4-lane MIPI D-PHY 1.2 interfaces
 - Two CSI-2 1.3 Rx controller(s)
 - 2.5Gbps/lane peak data rate
 - 4 virtual channel/controller support, Total 8 virtual channels
- 16-bit Parallel input, up to 1080p
- SLVS-EC and HiSPi sensor supported via external bridges

3.5 ISP

- 2 ISP core
- Each core runs at 400MHz, one-pixel/cycle pipeline
- Each ISP core supports up to 4 channel processing
- Each ISP core has 3 views output to DDR
 - View0 support YUV422/420 output
 - View1 support YUV422/420 output with downscale
 - View2 support Y channel output
- Adjustable AE and AWB

- Adjustable 33 knee point inversed PWL
- RGGB/RCCB/RYYB sensor support, 4C sensor support
- Lens shading correction, dynamic color shading correction
- Defect pixel correction
- Color correction
- Raw DNS and UV DNS
- Bayer CFA de-mosaicing
- Edge-adaptive sharpening
- Gamma correction and contrast adjustment
- Tone mapping
- Cropping, resizing
- Multi-exposure HDR
 - up to 3 exposures per stream
 - 120dB dynamic range
- Night vision enhancement
- LED flicker suppression

3.6 Video codec

The maximum encoding capability of video codec is 4*720p@25fps. The following codec standards are supported for encoding:

- H.264 Baseline Profile (BP).
- H.264 Main Profile (MP).
- H.264 High Profile (HP).

The maximum decoding capability of video codec is 4*720p@25fps. The following codec standards are supported for decoding:

- H.264 Baseline Profile (BP).
- H.264 Main Profile (MP).
- H.264 High Profile (HP).

3.7 VOUT Interface

- 24-bit RGB parallel VOUT, up to 1080p@60fps
 - EIA/CEA-861-B standard format
 - Up to 165MHz pixel clock
 - Multiple output modes
 - Support BT.656 style embedded synchronization
 - Support BT.601 discrete synchronization, support YUV422 output
 - Support RGB888 and YUV444 output
 - Support YUV420 or YUV422 input color formats

3.8 Computer Vision(CV)

Computer vision acceleration is realized using a combination of dedicated HW acceleration modules and Vision DSP. The following functions are implemented in dedicated HW acceleration module:

- GWARP: Affine/perspective transform
 - 800MHz running frequency
 - Support bilinear, bicubic interpolation

- Support RGB/YUV422/420 input/output
- LUT based warp engine
- Scaler: Upscale/Downscale acceleration engine
 - Support RGB/YUV422/420 input/output
 - Support polyphase and pyramid mode

In addition, there are two high-performance wide SIMD Vision DSP processors, each capable of performing 32-way or 64-way computations. They are used to support the following most commonly used CV kernels or functions:

- 2 Vision DSP running at 800MHz frequency
- Each DSP has 25.6 GFLOPS or 0.4TOPS
- Support OpenVX, eigen and other CV library for following APIs
 - Stereo vision
 - Optical flow/SfM
 - SLAM
 - Image pyramid
 - Feature extraction (HOG, Harris, ORB)

3.9 Neural Network(BSNN)

- Dedicated 800MHz DSP for control logic and custom operators
- HW based 3-D CNN MAC array computation
 - 3-D tensor MAC array with high utilization
 - Structured weight pruning and compression
 - Up-to 4x sparsity computation speed up
 - Fixed point 8-bit weight, 8-bit or 4-bit data precision
 - Per layer and per channel quantization
 - Non-power-of-2 quantization
- HW based 2-D GEMM engine
 - 512 MAC for int16 and FP16
 - 1024 MAC for int8
- HW based 1-D EDP engine
 - Element wise: add, subtract, mul, max, min, abs, neg, clamp, scale, cmp
 - Vector reduction: radd, rmax, rmin
 - Non-linear: tanh, sigmoid, exp, ln, reciprocal

3.10 Safety and Security

- Safety
 - 400MHz dual-core ARM Cortex-R5 safety processor configured as DCLS (dual-core lockstep)
 - Run-time functional and BIST self-test control
 - Watchdog timer
 - PVT monitor interface
 - XIP capable OSPI NOR flash controller
 - Safety-DMA for peripheral data transfer
- Security
 - 200MHz dedicated secure MCU

- Secure OTP and security boot ROM
- HW true random number generator
- Crypto modules: AES, SHA2, RSA/ECC (elliptic curve), SM2/3/4, CRC
- Security and security bus fabric
- Shared 1MB SRAM w. ECC protection and memory sand box

3.11 High-speed IO Interfaces

- 2x automotive Ethernet MAC interfaces
 - TSN and QoS support
 - Support 802.1Qav, 802.1AS, 802.1Qbv, 802.1bu
 - Tx and Rx DMA
 - Configurable MII, RGMII or GMII interfaces
 - MDIO master interface for external PHY control
- 1x USB 3.0 DRD interface + 1x USB 2.0 device interface
 - USB 3.0 host mode or device mode, compatible with 2.0 mode
 - USB 2.0 device mode
 - USB 3.0 Serdes PHY and USB 2.0 UTMI+ PHY
 - VBUS charging support
- 2x eMMC/SD interfaces
 - 8-bit eMMC 5.1 host
 - eMMC NAND flash
 - Compatibility with legacy MMC card
 - HS SDR 50Mhz
 - 4-bit SD 4.2 host
 - SDSC, SDHC, SDXC microSD card
 - Support DS/HS/SDR12/SDR25/SDR50
- PCIE Gen3 interface
 - 4-lane PCIE Gen3 SerDes PHY
 - Configurable to 2x2 or 1x4 modes via PHY bifurcation support
 - Dual PCIE Gen3 controller in RC or EP mode

3.12 DMA

- Independent core, slave interface clock, master interface clock is synchronous to the core clock
- Up to two channels, one per source and destination pair AMBA 4 AXI master interface
- AHB slave interface for programming the DMA controller
- Little Endian mode for AXI master interface and LLI access
- DMAC status indication outputs. Idle/busy indication
- DMA hold function
- Multiple levels of DMA transfer hierarchy
- DMA transfer split into transaction, block, and complete DMA transfer levels
- Support for AXI unaligned transfers

3.13 Peripheral Interfaces

Following peripheral interfaces can be used by Safety R5 or A55:

- 6x I2C (3x support SMBUS)

- 4x UART
- 6x SPI (4x master, 2x slave)
- 2x I2S (I2S0 master, I2S1 slave)
- 246x GPIO
- 4x PWM
- 3x CAN-FD
- 13x timer
- 13x watchdog

3.14 Package

900 balls, 25 × 25 mm, 0.8 mm pitch FCBGA

3.15 ISO26262 Information

Function safety support up to ASIL B

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4 Electrical characteristics

4.1 Chip-level conditions

4.1.1 Absolute maximum ratings

Stresses beyond those listed under Table 4-1 may affect reliability or cause permanent damage to the device. These are stress ratings only. Functional operation of the device at these or any other conditions beyond those indicated in the operating ranges or parameters tables is not implied.

Table 4-1 Absolute maximum ratings

Parameter description	Symbol	Min	Max	Unit	Notes
Core supply voltages	VDD	0	0.88	V	
	VDD_DDR0_MIPI	0	0.88	V	
	VDD_DDR1	0	0.88	V	
	VDD_GPU	0	0.88	V	
	VDD_ISP_CV	0	0.88	V	
	VDD_NET	0	0.88	V	
	VDD_VSP	0	0.88	V	
Power supply for GPIO	VDDIO_AON	0	1.98	V	
	VDDIO_SOC	0	1.98	V	
Power supply for SDIO	VDDIO_SDIO_0	0	3.30(+5%)	V	
	VDDIO_SDIO_1	0	3.30(+5%)	V	
Power supply for USB	USB20_DVDD	0	0.80(+10%)	V	
	USB20_VDD33	0	3.30(+10%)	V	share with USB20_VDDH USB20_VPH on ball
	USB20_VP	0	0.80(+10%)	V	
	USB20_VPTX	0	0.80(+10%)	V	
	USB30_DVDD	0	0.80(+10%)	V	
	USB30_VDD33	0	3.30(+10%)	V	share with USB20_VDDH USB20_VPH on ball
	USB30_VP	0	0.80(+16.875%)	V	
	USB30_VPTX	0	0.80(+10%)	V	
Power supply for PCIE	PCIE0_VP	0	0.80(+10%)	V	share with PCIE0_VPTX, PCIE0_VPDIG on ball
	PCIE0_VPH	0	1.80(+10%)	V	
	PCIE1_VP	0	0.80(+10%)	V	share with PCIE0_VPTX, PCIE0_VPDIG on ball
	PCIE1_VPH	0	1.80(+10%)	V	
Power supply for DDR	DDR0_VAA	0	1.80(+10%)	V	
	DDR0_VDDQ	0	1.17	V	
	DDR1_VAA	0	1.80(+10%)	V	
	DDR1_VDDQ	0	1.17	V	
Power supply for MIPI	MIPI_VP	0	0.88	V	4 MIPI blocks, total 6 MIPI PHYs share on ball
	MIPI_VPH	0	1.98	V	4 MIPI blocks, total 6

					MIPI PHYs share on ball
Power supply for PLL	PLL_AON_VDDHV	0	1.98	V	
	PLL_DDR0_VDDHV	0	1.98	V	
	PLL_DDR1_VDDHV	0	1.98	V	
	PLL_PCIE_VDD	0	0.88	V	
	PLL_PCIE_VDDHV	0	1.98	V	
	PLL_TOP_VDD	0	0.88	V	
	PLL_TOP_VDDHV	0	1.98	V	
Power supply for PVT	PVT_VDDA_1	0	1.80(+10%)	V	
	PVT_VDDA_2	0	1.80(+10%)	V	
	PVT_VDDA_3	0	1.80(+10%)	V	
	PVT_VDDA_4	0	1.80(+10%)	V	
	PVT_VDDA_5	0	1.80(+10%)	V	
Power supply for OTP	OTP_VDD2	0	1.98	V	
Input/output voltage range	Vin/Vout			V	
ESD damage immunity:				V	
• Human Body Model (HBM)		-	2000	V	
• Charged Device Model (CDM)		-	5	A	
Storage temperature range	T_STORAGE	-40	125	°C	

4.1.2 Thermal resistance

Table 4-2 provides the thermal resistance characteristics for the package used on this device. Natural convection (Four-layer board 2s2p).

Table 4-2 Thermal Resistance Characteristics

PARAMETER	DESCRIPTION	°C/W
R _{θJC}	Junction to case	0.29
R _{θJB}	Junction to board	2.62
R _{θJA}	Junction to free air	6.99
	Junction to moving air	6.32 ¹
Ψ _{JT}	Junction to package top	0.15
Ψ _{JB}	Junction to board	2.20

Note: 1. The air flow is 2m/s.

4.1.3 Operating ranges

Table 4-3 provides the operating ranges of the A1000L processor. For details on the chip's power structure, see the “Power Management Unit (PMU)” chapter of *A1000 Reference Manual*.

Table 4-3 Operating ranges

Parameter description	Symbol	Min	Typ	Max	Unit	Notes
Core supply voltages	VDD	0.77	0.80	0.88	V	
	VDD_DDR0_MIPI	0.77	0.80	0.88	V	
	VDD_DDR1	0.77	0.80	0.88	V	
	VDD_GPU	0.77	0.80	0.88	V	
	VDD_ISP_CV	0.77	0.80	0.88	V	
	VDD_NET	0.77	0.80	0.88	V	
	VDD_VSP	0.77	0.80	0.88	V	
Power supply for GPIO	VDDIO_AON	1.80(-7%)	1.80	1.80(+10%)	V	
	VDDIO_SOC	1.80(-7%)	1.80	1.80(+10%)	V	

Power supply for SDIO	VDDIO_SDIO_0	1.67/2.80	1.80/3.00	1.98/3.30	V	
	VDDIO_SDIO_1	1.67/2.80	1.80/3.00	1.98/3.30	V	
Power supply for USB	USB20_DVDD	0.80(-6%)	0.80	0.80(+10%)	V	
	USB20_VDD33	3.30(-6%)	3.30	3.30(+10%)	V	Share with USB20_VDDH USB20_VPH on ball
	USB20_VP	0.80(-6%)	0.80	0.80(+10%)	V	
	USB20_VPTX	0.80(-6%)	0.80	0.80(+10%)	V	
	USB30_DVDD	0.80(-6%)	0.80	0.80(+10%)	V	
	USB30_VDD33	3.30(-6%)	3.30	3.30(+10%)	V	Share with USB20_VDDH USB20_VPH on ball
	USB30_VP	0.80(-6%)	0.80	0.80(+10%)	V	
	USB30_VPTX	0.80(-6%)	0.80	0.80(+10%)	V	
Power supply for PCIE	PCIE0_VP	0.80(-6%)	0.80	0.80(+10%)	V	Share with PCIE0_VPTX, PCIE0_VPDIG on ball
	PCIE0_VPH	1.80(-5%)	1.80	1.80(+10%)	V	
	PCIE1_VP	0.80(-6%)	0.80	0.80(+10%)	V	share with PCIE0_VPTX, PCIE0_VPDIG on ball
	PCIE1_VPH	1.80(-5%)	1.80	1.80(+10%)	V	
Power supply for DDR	DDR0_VAA	1.80(-7%)	1.80	1.80(+10%)	V	
	DDR0_VDDQ	1.07	1.10	1.17	V	
	DDR1_VAA	1.80(-7%)	1.80	1.80(+10%)	V	
	DDR1_VDDQ	1.07	1.10	1.17	V	
Power supply for MIPI	MIPI_VP	0.80(-6%)	0.80	0.88	V	4 MIPI blocks, total 6 MIPI PHYs share on ball
	MIPI_VPH	1.80(-6%)	1.80	1.98	V	4 MIPI blocks, total 6 MIPI PHYs share on ball
Power supply for PLL	PLL_AON_VDDHV	1.62	1.80	1.98	V	
	PLL_DDR0_VDDHV	1.62	1.80	1.98	V	
	PLL_DDR1_VDDHV	1.62	1.80	1.98	V	
	PLL_PCIE_VDD	0.77	0.80	0.88	V	
	PLL_PCIE_VDDHV	1.62	1.80	1.98	V	
	PLL_TOP_VDD	0.77	0.80	0.88	V	
	PLL_TOP_VDDHV	1.62	1.80	1.98	V	
Power supply for PVT	PVT_VDDA_1	1.80(-10%)	1.80	1.80(+10%)	V	
	PVT_VDDA_2	1.80(-10%)	1.80	1.80(+10%)	V	
	PVT_VDDA_3	1.80(-10%)	1.80	1.80(+10%)	V	
	PVT_VDDA_4	1.80(-10%)	1.80	1.80(+10%)	V	
	PVT_VDDA_5	1.80(-10%)	1.80	1.80(+10%)	V	
Power supply for OTP	OTP_VDD2	1.80(-10%)	1.80	1.80(+10%)	V	
Temperature sensor accuracy	Tdelta	-1	-	1	°C	The trimming result
Junction temperature, industrial	Tj	-40	-	125	°C	

Note: The power design should according to the typical values.

4.1.4 External clock sources

A 25 MHz clock “XTAL_IN” is needed as the primary clock source for the PLLs to generate the clock for A1000L. Another 25Mhz clock “XTAL_INB” is optional, it can be used as the primary clock source for the PLLs for CPU, BUS, and high-speed interface, it can be monitored by

“XTAL_IN”.

- Reference clock stability need be less than 50ppm
- The DC parameters of “XTAL_IN” and “XTAL_INB” are listed in Table 4-16 and Table 4-17

Note: the "XTAL_INB" cannot be used as reference clock for safety subsystem PLLs.

4.1.5 Maximum supply currents

Power consumption is highly dependent on the application. Estimating the maximum supply currents required for power supply design is difficult because the use cases that requires maximum supply current is not a realistic use case. The results provided are intended to be used as guidelines for power supply design.

Table 4-4 Maximum supply currents

Power Rail	Normal DC Voltage / V	Max DC Current / mA	Max Power / mW
VDD	0.8	12347	9878
VDD_DDR0_MIPI	0.8	1297	1037
VDD_DDR1	0.8	1277	1021
VDD_GPU	0.8	1028	822
VDD_ISP_CV	0.8	3097	2477
VDD_NET	0.8	12257	9806
VDD_VSP	0.8	521	417
VDDIO_AON	1.8	400	720
VDDIO_SDIO_0	1.8/3.0	150	270
VDDIO_SDIO_1	1.8/3.0	150	270
VDDIO_SOC	1.8	550	990
DDR0_VAA	1.8	10	18
DDR0_VDDQ	1.1	716	788
DDR1_VAA	1.8	10	18
DDR1_VDDQ	1.1	716	788
MIPI_VP	0.8	70	56
MIPI_VPH	1.8	30	54
OTP_VDD2	1.8	50	90
PCIE0_VP	0.8	200	160
PCIE0_VPH	1.8	150	270
PCIE1_VP	0.8	200	160
PCIE1_VPH	1.8	150	270
PLL_AON_VDDHV	1.8	20	36
PLL_DDR0_VDDHV	1.8	20	36
PLL_DDR1_VDDHV	1.8	20	36
PLL_PCIE_VDD	0.8	10	8
PLL_PCIE_VDDHV	1.8	20	36
PLL_TOP_VDD	0.8	150	120

PLL_TOP_VDDHV	1.8	90	162
PVT_VDDA_1	1.8	10	18
PVT_VDDA_2	1.8	10	18
PVT_VDDA_3	1.8	10	18
PVT_VDDA_4	1.8	10	18
PVT_VDDA_5	1.8	10	18
USB20_DVDD	0.8	20	16
USB20_VDD33	3.3	120	396
USB20_VPTX	0.8	30	24
USB30_DVDD	0.8	20	16
USB30_VDD33	3.3	120	396
USB30_VPTX	0.8	30	24
USB20_VP	0.8	80	64
USB30_VP	0.8	80	64

Note: the Max DC Current is the max avg current value, not the peak value.

4.1.6 Power modes

Table 4-5 Power domain

Power Domain	Power rail	Description
AON domain	VDD	always on
ISP_CV domain	VDD_ISP_CV	2 DSP core and 2 ISP core
GPU domain	VDD_GPU	GPU
NET domain	VDD_NET	whole NET
VSP domain	VDD_VSP	Whole VSP

4.1.7 USB PHY DC electrical characteristics

The USB 3.0 femtoPHY is fully compliant with the USB3.0 SuperSpeed Transmitter and Receiver Normative Electrical Parameters as defined in Universal Serial Bus 3.0 Specification, Revision 1.0.

Supports 5.0-Gbps SuperSpeed (SS) through 3-m cable, 480-Mbps high-speed (HS), 12-Mbps fullspeed (FS), and 1.5-Mbps low-speed (LS) serial (Host mode only) data transmission rates, with a high-speed (HS) transmit-to-receive interpacket delay requirement of 38 bit-times as implied by the UTMI specification, version 1.05.

The following table provides the worst-case maximum current specifications that the USB 3.0 femtoPHY power supplies must be able to handle.

Table 4-6 Worst-Case Maximum Current

vph<#>	vp	vptx<#>	VDD33<#>	VDDH<#>	DVDD	Unit
25.7	36.1	12.6	19.6	13.02	9.8	mA

The following table provides the fastest ramp-up times allowed for the USB 3.0 femtoPHY power supplies without falsely triggering the ESD circuitry.

Table 4-7 Maximum Power Supply Slew Rates

vph<#>	vp	vptx<#>	VDD33<#>	VDDH<#>	DVDD	Unit
0.1	0.1	0.1	0.1	0.1	0.1	V/us

4.1.8 PCIe PHY DC electrical characteristics

The PCIe PHY complies with the required features specified in: Peripheral Component Interconnect Express 3.0 specification, Revision 1.0.

- PCI Express Base Specification, Revision 3.1, November 2013, PCI-SIG
- PCI Express Card Electromechanical Specification, Revision 3.0, July 2013, PCI-SIG
- PCI Express Mini Card Electromechanical Specification, Revision 2.0, April 2012, PCI-SIG
- IEEE Standard 1149.1-2001, IEEE
- IEEE Standard 1149.6-2003, IEEE

The PHY supports the power supply voltages in Table 3-6. The values are supported across a temperature range of -40°C to 125°C.

Table 4-8 Supported Power Supply Limits at the PHY Macro Bumps

Supply	Description	Min	Typical	Max	Units
PCIE_VPH	High Voltage IO supply	1.8-5%	1.8	1.8+10%	V
PCIE_VP	Analog supply	0.8-7%	0.8	0.8+10%	V

4.1.9 MIPI DPHY DC electrical characteristics

The MIPI DPHY is conformant with the MIPI D-PHY interface specification, revision v1.2 and backwards compatible with previous D-PHY specification version 1.1.

Table 4-9 Input DC Specifications (Apply to CLKP/N and DATAP/N Inputs)

Symbol	Parameter	Minimum	Maximum	Unit	Notes
VPIN	Pin signal voltage range	-50	1350	mV	
ILEAK	Pin leakage current	-10	10	μA	$VGND_{SH}(\min) \leq VPIN \leq VGND_{SH}(\max) + V_{OH}(\text{absmax})$ Lane module in LP receive mode
VGND _{SH}	Ground shift	-50	50	mV	
VPIN(absmax)	Maximum transient output voltage level	-0.15	1.45	V	
VPIN(absmax)	Maximum transient time above $V_{OH}(\text{absmax})$		20	ns	The voltage overshoot and undershoot beyond the VPIN is only allowed during a single 20ns window after any LP-0 to LP-1 transition or vice versa. For all other situations the voltage overshoot and undershoot must stay within the VPIN range.

Table 4-10 HS Line Receiver DC Specifications

Symbol	Parameter	Mini	Typical	Max	Unit	Notes
VIDTH	Differential input high voltage threshold			70	mV	D-PHY spec 1.1 compatibility mode ($\leq 1.5\text{Gbps}$)
				40	mV	In case of High-Speed deskew calibration ($> 1.5\text{Gbps}$)
VIDTL	Differential input low voltage threshold	-70			mV	D-PHY spec 1.1 compatibility mode ($\leq 1.5\text{Gbps}$)
		-40			mV	In case of High-Speed deskew calibration ($> 1.5\text{Gbps}$)
VIHHS	Single ended input high voltage			460	mV	Excluding possible additional RF interference of 100mV peak sine wave beyond 450MHz.
VILHS	Single ended input low voltage	-40			mV	Excluding possible additional RF interference

						of 100mV peak sine wave beyond 450MHz.
VCMRXDC	Input common mode voltage	70		330	mV	Excluding possible additional RF interference of 100mV peak sine wave beyond 450MHz. This table value includes a ground difference of 50mV between the transmitter and the receiver, the static common-mode level tolerance and variations below 450MHz
ZID	Differential input impedance	80	100	125	Ω	

Table 4-11 LP Line Receiver DC Specifications

Symbol	Parameter	Minimum	Typical	Maximum	Unit
VIL	Input low voltage			550	mV
VIH	Input high voltage	740			mV
VIL-ULPS	Logic 0 input voltage, ULPS			300	mV
VHYST	Input hysteresis	25			mV

4.1.10 DDR**4.1.10.1 Operating Conditions**

All voltages are referenced to VSS.

4.1.10.2 Recommended Operating Conditions

The following table shows the DDR operating conditions, including supply voltages and temperature.

Table 4-12 Recommended Operating Conditions¹

Symbol	Description	Minimum	Typical	Maximum	Units
VDD ^{2 3 4}	Core Voltage Supports data rates of 3200 Mbps and below	V(Typ) -7%	0.80	V(Typ) +10%	V
VRIP_VDD	Core power/ground supply rail ripple: LPDDR4 mode	-2.0		2.0	% VDD
VDDQ ²	VDDQ supply (LPDDR4), DC average ⁵	1.06	1.1	1.17	V
VRIP_VDDQ	IO power/ground supply rail ripple, LPDDR4 mode	-2.5		2.5	% VDDQ
VAA ²	PLL supply voltage connected to independent 1.8 V supply VAA	VAA(Typ) -7%	1.80	VAA(Typ) +10%	V
VRIP_PLL	PLL power/ground supply rail when connected to VAA(Typ) = 1.8 V	-2.5		2.5	%VAA (VDDQ)
RVDDQ RVAA	Supply power-up ramp rate for VDDQ and VAA ⁶			5	mV/us
TJ	Junction temperature	-40		125	°C

1. Correct operation of the DDR PHY is not guaranteed if the recommended operating conditions for VAA, VDDQ, VDD are exceeded.
2. Voltages VDD, VAA and VDDQ are measured at the pins of the hard DDR PHY macrocells.
3. Core voltage range includes noise.
4. Power and jitter analysis is performed at DC average VDD levels of +/-5%.
5. VDDQ voltage range is for DC voltage only. DC is inclusive of all noise up to 20 Mhz.
6. The VDDQ and VAA power supplies are limited to a maximum supply ramp rate listed in Table 4-12 above. This limit prevents triggering of the internal clamps on those power

supplies. All designs must make a trade-off around protecting from ESD events. A super-fast power rail ramp rate during normal operation looks significantly like an ESD discharge that has the potential to damage the chip. Our power rail requirements are linked to our trade-offs between allowing power rails to be driven by a wide variety of sources with different characteristics, versus having ESD clamps which trigger at conservative rates to ensure the chip is not damaged.

4.1.10.3 Core Voltage Tolerance

The following figures show the core voltage tolerance, allowable jitter, and relationship to PVT points for both minimum and maximum voltages.

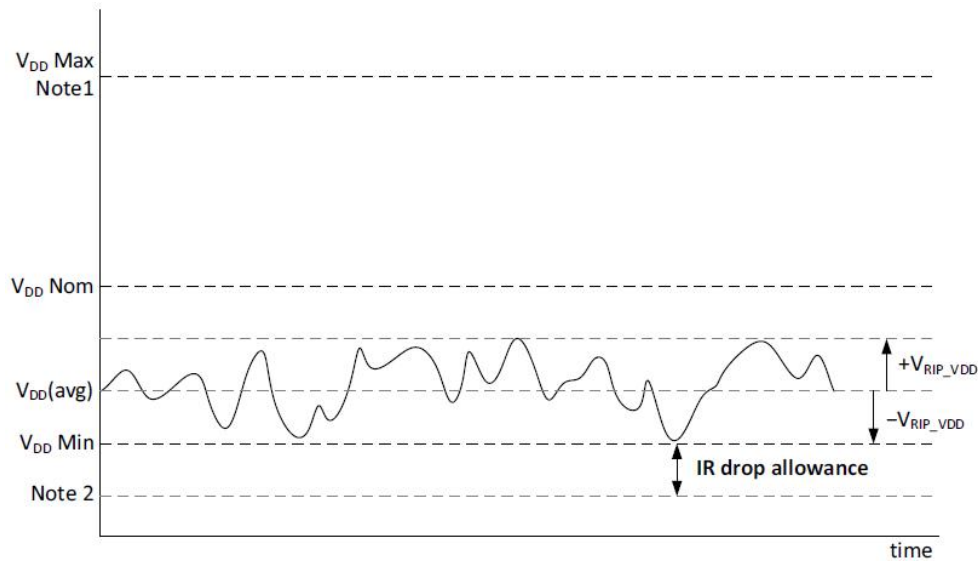


Figure 4-1 Core Voltage Tolerance for Minimum VDD Voltage

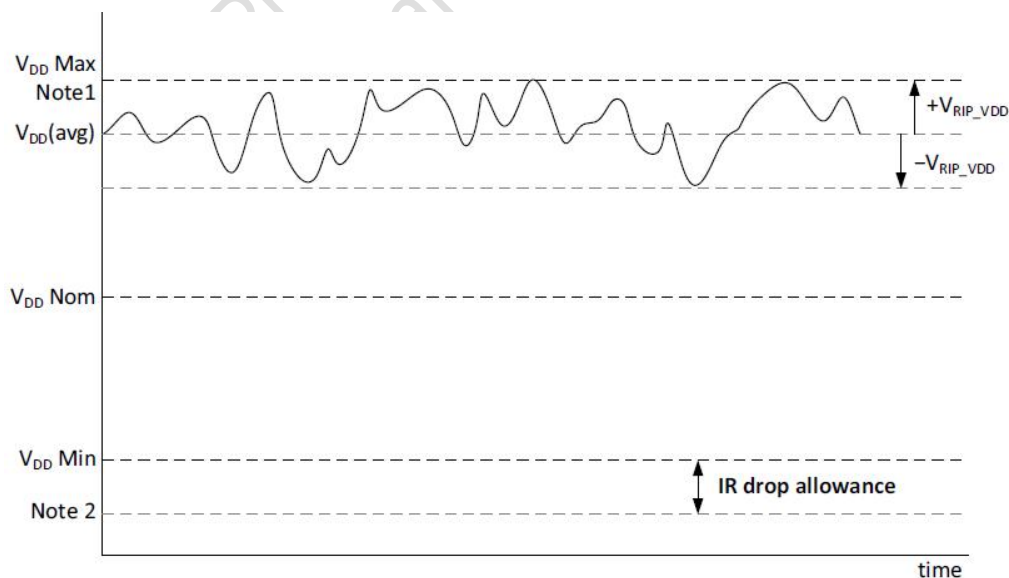


Figure 4-2 Core Voltage Tolerance for Maximum VDD Voltage

Figure Notes:

1. VDDMax is the maximum voltage
2. As shown in the figures, there is an IR drop allowance for the supply voltage to the

individual transistors in A1000L die between VDDMin and the minimum voltage.

4.2 Power supplies requirements and restrictions

4.2.1 Power-up sequence

Table 4-13 A Power-up Sequence Example

Power Rail	Voltage (Unit:V)	Recommend Power Up Sequence			Power Up Sequence Constraints
		t0	t0+2ms	t0+4ms	
VDD	0.8				VDD
VDD_DDR0_MIPI	0.8				same with VDD
VDD_DDR1	0.8				same with VDD
VDD_GPU	0.8				same with VDD
VDD_ISP_CV	0.8				same with VDD
VDD_NET	0.8				same with VDD
VDD_VSP	0.8				same with VDD
VDDIO_AON	1.8				ramp up before VDD
VDDIO_SOC	1.8				ramp up before VDD
VDDIO_SDIO_0	1.8/3.0				ramp up before VDD
VDDIO_SDIO_1	1.8/3.0				ramp up before VDD
USB20_DVDD	0.8				same with VDD
USB20_VDD33	3.3				rule 0: USB20_VDD33 can ramp up before USB20_VPTX/USB20_DVDD , but must not exceed 10ms
					rule 1: USB20_VDD33 can ramp up same with USB20_VPTX/USB20_DVDD
					rule 2: USB20_VDD33 can ramp up later than USB20_VPTX/USB20_DVDD
USB20_VP	0.8				same with VDD
USB20_VPTX	0.8				same with VDD
USB30_DVDD	0.8				same with VDD
USB30_VDD33	3.3				rule 0: USB30_VDD33 can ramp up before USB30_VPTX/USB30_DVDD , but must not exceed 10ms
					rule 1: USB30_VDD33 can ramp up same with USB20_VPTX/USB20_DVDD
					rule 2: USB30_VDD33 can ramp up later than USB30_VPTX/USB30_DVDD
USB30_VP	0.8				same with VDD
USB30_VPTX	0.8				same with VDD
PCIE0_VP	0.8				same with VDD
PCIE0_VPH	1.8				no requirement
PCIE1_VP	0.8				same with VDD

PCIE1_VPH	1.8			no requirement
DDR0_VAA	1.8			no requirement
DDR0_VDDQ	1.1			no requirement
DDR1_VAA	1.8			no requirement
DDR1_VDDQ	1.1			no requirement
MIPI_VP	0.8			not before VDD
MIPI_VPH	1.8			later than VDD
PLL_AON_VDDHV	1.8			no requirement
PLL_DDR0_VDDHV	1.8			no requirement
PLL_DDR1_VDDHV	1.8			no requirement
PLL_PCIE_VDD	0.8			same with VDD
PLL_PCIE_VDDHV	1.8			no requirement
PLL_TOP_VDD	0.8			same with VDD
PLL_TOP_VDDHV	1.8			no requirement
PVT_VDDA_1	1.8			no requirement
PVT_VDDA_2	1.8			no requirement
PVT_VDDA_3	1.8			no requirement
PVT_VDDA_4	1.8			no requirement
PVT_VDDA_5	1.8			no requirement
OTP_VDD2	1.8			VDD power up first, then wait for large than 10ns, then power up OTP_VDD2

Note: RST_N need keep low at least 3ms after all power up

4.2.2 Power-down sequence

Preferred:

Turn off the VDDIO_SOC/VDDIO_AON/ VDDIO_SDIO_0/VDDIO_SDIO_1 last.

Acceptable:

Turn off all powers at the same time.

Not Recommended:

Turn off the VDDIO_SOC/VDDIO_AON/VDDIO_SDIO_0/VDDIO_SDIO_1 first.

Note: RST_N must be low before power off.

4.2.3 AC noise requirement

Table 4-14 AC noise

Power rail	Voltage (Unit:V)	AC noise
VDD	0.8	+2.5%;-2.5%
VDD_DDR0_MIPI	0.8	+2%;-2%
VDD_DDR1	0.8	+2%;-2%
VDD_GPU	0.8	+2.5%;-2.5%
VDD_ISP_CV	0.8	+2.5%;-2.5%
VDD_NET	0.8	+4.5%;-4.5%
VDD_VSP	0.8	+2.5%;-2.5%
VDDIO_AON	1.8	+3%;-3%
VDDIO_SOC	1.8	+3%;-3%

VDDIO_SDIO_0	1.8/3.0	+3%;-3%
VDDIO_SDIO_1	1.8/3.0	+3%;-3%
USB20_DVDD	0.8	+3%;-3%
USB20_VDD33	3.3	+3%;-3%
USB20_VP	0.8	+3%;-3%
USB20_VPTX	0.8	+3%;-3%
USB30_DVDD	0.8	+3%;-3%
USB30_VDD33	3.3	+3%;-3%
USB30_VP	0.8	+3%;-3%
USB30_VPTX	0.8	+3%;-3%
PCIE0_VP	0.8	+5%;-5%
PCIE0_VPH	1.8	+3%;-3%
PCIE1_VP	0.8	+5%;-5%
PCIE1_VPH	1.8	+3%;-3%
DDR0_VAA	1.8	+2.5%;-2.5%
DDR0_VDDQ	1.1	+2.5%;-2.5%
DDR1_VAA	1.8	+2.5%;-2.5%
DDR1_VDDQ	1.1	+2.5%;-2.5%
MIPI_VP	0.8	+3%;-3%
MIPI_VPH	1.8	+3%;-3%
PLL_AON_VDDHV	1.8	+10mv;-10mv
PLL_DDR0_VDDHV	1.8	+10mv;-10mv
PLL_DDR1_VDDHV	1.8	+10mv;-10mv
PLL_PCIE_VDD	0.8	+10mv;-10mv
PLL_PCIE_VDDHV	1.8	+10mv;-10mv
PLL_TOP_VDD	0.8	+10mv;-10mv
PLL_TOP_VDDHV	1.8	+10mv;-10mv
PVT_VDDA_1	1.8	+20mv;-20mv
PVT_VDDA_2	1.8	+20mv;-20mv
PVT_VDDA_3	1.8	+20mv;-20mv
PVT_VDDA_4	1.8	+20mv;-20mv
PVT_VDDA_5	1.8	+20mv;-20mv
OTP_VDD2	1.8	+20mv;-20mv

4.3 PLL electrical characteristics

Table 4-15 PLL electrical characteristics

PLL Type	Parameter	Value
PLL_CPU	Clock output range	16MHZ-3200MHZ
	Reference clock	25MHZ
	Lock time	10us
	Period jitter	21.7 ps
PLL_DSU	Clock output range	1300MHZ
	Reference clock	25MHZ
	Lock time	10us
	Period jitter	25.3ps
PLL_SYSBUS	Clock output range	16MHZ-3200MHZ
	Reference clock	25MHZ
	Lock time	10us

	Period jitter	36.5ps
PLL_COREIP	Clock output range	16MHZ-3200MHZ
	Reference clock	25MHZ
	Lock time	10us
	Period jitter	36.5ps
PLL_DISPLAY	Clock output range	16MHZ-3200MHZ
	Reference clock	25MHZ
	Lock time	10us
	Period jitter	19.4ps
PLL_HSPLSP	Clock output range	16MHZ-3200MHZ
	Reference clock	25MHZ
	Lock time	10us
	LTJ jitter	6ps RMS in 100KHZ-10MHZ
PLL_DDR	Clock output range	16MHZ-1066MHZ
	Reference clock	25MHZ
	Lock time	10us
	Period jitter	36.5ps
PLL_main	Clock output range	16MHZ-3200MHZ
	Reference clock	25MHZ
	Lock time	10us
	Period jitter	0.735ps
PLL_LSP	Clock output range	16MHZ-3200MHZ
	Reference clock	25MHZ
	Lock time	10us
	Period jitter	0.941ps

4.4 I/O DC parameters

This section includes the DC parameters of the following I/O types:

- General Purpose I/O (GPIO)DC parameters
- SDIO DC parameters
- DDR I/O DC parameters

4.4.1 General purpose I/O (GPIO) DC parameters

4.4.1.1 Recommended Operating Conditions



Warning: Permanent damage could occur if the operation exceeds the ranges listed in Table 4-16.

Table 4-16 Recommended Operating Conditions

Parameter		Min.	Nom.	Max.	Units
VDD	Pre-Driver Voltage	0.72	0.8	0.88	V
VDDPST	Post-Driver Voltage	1.62	1.8	1.98	V
VIMAX	Maximum Input Voltage			1.98	V
VDDramp-upslew	Ramp up slew for VDD			0.018	V/us
VDDIO ramp-upslew	Ramp up slew for VDDIO			0.018	V/us

Table 4-17 DC Characteristics

Parameter		Min.	Nom.	Max.	Units
V _{IL}	Input Low Voltage	-0.3		0.35* VDDIO	V

V_{IH}	Input High Voltage	0.65* VDDIO		1.98	V
V_T	Threshold Point	0.82	0.92	1	V
V_{T+}	Schmitt Trigger Low to High Threshold Point	0.98	1.1	1.21	V
V_{T-}	Schmitt Trigger High to Low Threshold Point	0.74	0.82	0.9	V
V_{TPU}	Threshold Point with Pull-up Resistor Enabled	0.82	0.91	1	V
V_{TPD}	Threshold Point with Pull-down Resistor Enabled	0.83	0.93	1.02	V
V_{T+}	Schmitt Trigger Low to High Threshold Point with Pull-up Resistor Enabled	0.97	1.09	1.2	V
V_{T-}	Schmitt Trigger High to Low Threshold Point with Pull-up Resistor Enabled	0.73	0.81	0.89	V
V_{T+}	Schmitt Trigger Low to High Threshold Point with Pull-down Resistor Enabled	0.99	1.11	1.22	V
V_{T-}	Schmitt Trigger High to Low Threshold Point with Pull-down Resistor Enabled	0.75	0.83	0.91	V
I_I I_{OZ}	Input Leakage Current @ $V_I = 1.8V$ or 0V Tri-state Output Leakage Current @ $V_O = 1.8V$ or 0V			$\pm 10\mu$ $\pm 10\mu$	A A
R_{PU}	Pull-up Resistor	17k	24k	40k	Ω
R_{PD}	Pull-down Resistor	17k	23k	36k	Ω
V_{OL}	Output Low Voltage			0.45	V
V_{OH}	Output High Voltage	1.35			V
I_{OL}	Low Level Output Current @ $V_{OL}(\max)$				
	(DS3,DS2,DS1,DS0) = '0000'	2.0	3.0	4.0	mA
	(DS3,DS2,DS1,DS0) = '0001'	4.2	6.5	8.6	mA
	(DS3,DS2,DS1,DS0) = '0010'	6.0	9.3	19.7	mA
	(DS3,DS2,DS1,DS0) = '0011'	8.1	12.5	25.2	mA
	(DS3,DS2,DS1,DS0) = '0100'	11.2	17.4	31.2	mA
	(DS3,DS2,DS1,DS0) = '0101'	13.0	20.3	35.9	mA
	(DS3,DS2,DS1,DS0) = '0110'	14.5	22.6	40.0	mA
	(DS3,DS2,DS1,DS0) = '0111'	16.1	25.2	43.9	mA
	(DS3,DS2,DS1,DS0) = '1000'	21.3	33.5	49.3	mA
	(DS3,DS2,DS1,DS0) = '1001'	22.5	35.5	52.1	mA
	(DS3,DS2,DS1,DS0) = '1010'	23.4	36.9	54.9	mA
	(DS3,DS2,DS1,DS0) = '1011'	24.6	38.8	57.2	mA
	(DS3,DS2,DS1,DS0) = '1100'	26.2	41.1	59.5	mA
	(DS3,DS2,DS1,DS0) = '1101'	27.2	42.8	61.7	mA
	(DS3,DS2,DS1,DS0) = '1110'	27.9	43.9	63.5	mA
	(DS3,DS2,DS1,DS0) = '1111'	28.8	45.4	65.4	mA
I_{OH}	High Level Output Current @ $V_{OH}(\min)$				
	(DS3,DS2,DS1,DS0) = '0000'	1.7	2.8	3.9	mA
	(DS3,DS2,DS1,DS0) = '0001'	3.7	6.1	8.4	mA
	(DS3,DS2,DS1,DS0) = '0010'	5.3	8.7	12.0	mA
	(DS3,DS2,DS1,DS0) = '0011'	7.1	11.7	24.4	mA

	(DS3,DS2,DS1,DS0) = '0100'	9.8	16.2	30.2	mA
	(DS3,DS2,DS1,DS0) = '0101'	11.3	18.9	34.8	mA
	(DS3,DS2,DS1,DS0) = '0110'	12.5	20.9	38.8	mA
	(DS3,DS2,DS1,DS0) = '0111'	13.9	23.3	42.6	mA
	(DS3,DS2,DS1,DS0) = '1000'	18.3	30.9	47.9	mA
	(DS3,DS2,DS1,DS0) = '1001'	19.4	32.8	50.6	mA
	(DS3,DS2,DS1,DS0) = '1010'	20.2	34.1	53.3	mA
	(DS3,DS2,DS1,DS0) = '1011'	21.2	35.8	55.6	mA
	(DS3,DS2,DS1,DS0) = '1100'	22.5	38.0	57.8	mA
	(DS3,DS2,DS1,DS0) = '1101'	23.4	39.5	59.8	mA
	(DS3,DS2,DS1,DS0) = '1110'	24.0	40.6	61.6	mA
	(DS3,DS2,DS1,DS0) = '1111'	24.8	42.0	63.4	mA

Note: the default drive configuration of (DS3,DS2,DS1,DS0) is "0010"

4.4.2 SDIO DC parameters

4.4.2.1 Recommended Operating Conditions



Warning: Permanent damage could occur if the operation exceeds the ranges listed in Table 4-18.

Table 4-18 Recommended Operating Conditions

Parameter		Min.	Nom.	Max.	Units
VDD	Pre-Driver Voltage	0.72	0.8	0.88	V
VDDPST	Post-Driver Voltage	2.7	3	3.465	V
VIMAX	Maximum Input Voltage			3.465	V
VDDramp-upslew	Ramp up slew for VDD			0.018	V/us
VDDIORamp-upslew	Ramp up slew for VDDIO			0.018	V/us

Table 4-19 DC Characteristics

Parameter		Min.	Nom.	Max.	Units
V _{IL}	Input Low Voltage	-0.3		0.25* VDDIO	V
V _{IH}	Input High Voltage	0.625* VDDIO		1.98	V
V _T	Threshold Point	0.87	1.04	1.3	V
V _{T+}	Schmitt Trigger Low to High Threshold Point	1.0	1.15	1.4	V
V _{T-}	Schmitt Trigger High to Low Threshold Point	0.72	0.88	1.13	V
V _{TPU}	Threshold Point with Pull-up Resistor Enabled	0.86	1.02	1.28	V
V _{TPD}	Threshold Point with Pull-down Resistor Enabled	0.88	1.05	1.31	V
V _{T+}	Schmitt Trigger Low to High Threshold Point with Pull-up Resistor Enabled	0.99	1.14	1.39	V
V _{T-}	Schmitt Trigger High to Low Threshold Point with Pull-up Resistor Enabled	0.71	0.87	1.11	V
V _{T+}	Schmitt Trigger Low to High Threshold	1.01	1.16	1.42	V

	Point with Pull-down Resistor Enabled				
V_{T^-}	Schmitt Trigger High to Low Threshold Point with Pull-down Resistor Enabled	0.73	0.89	1.14	V
I_I I_{OZ}	Input Leakage Current @ $V_I = 1.8V$ or 0V Tri-state Output Leakage Current @ $V_O = 1.8V$ or 0V			$\pm 10\mu$ $\pm 10\mu$	A A
R_{PU}	Pull-up Resistor	31k	42k	67k	Ω
R_{PD}	Pull-down Resistor	24k	29k	33k	Ω
V_{OL}	Output Low Voltage			0.125* VDDIO	V
V_{OH}	Output High Voltage	0.75* VDDIO			V
I_{OL}	Low Level Output Current @ $V_{OL}(\max)$				
	(DS3,DS2,DS1,DS0) = '0000'	2.5	4.3	7.1	mA
	(DS3,DS2,DS1,DS0) = '0001'	3.8	6.5	10.7	mA
	(DS3,DS2,DS1,DS0) = '0010'	5.0	8.6	14.2	mA
	(DS3,DS2,DS1,DS0) = '0011'	6.3	10.8	17.7	mA
	(DS3,DS2,DS1,DS0) = '0100'	7.5	12.9	21.3	mA
	(DS3,DS2,DS1,DS0) = '0101'	8.8	15.1	24.9	mA
	(DS3,DS2,DS1,DS0) = '0110'	10.0	17.2	28.4	mA
	(DS3,DS2,DS1,DS0) = '0111'	11.2	19.3	31.9	mA
	(DS3,DS2,DS1,DS0) = '1000'	12.5	21.5	35.4	mA
	(DS3,DS2,DS1,DS0) = '1001'	13.8	23.6	39.0	mA
	(DS3,DS2,DS1,DS0) = '1010'	15.0	25.8	42.5	mA
	(DS3,DS2,DS1,DS0) = '1011'	16.2	27.9	46.0	mA
	(DS3,DS2,DS1,DS0) = '1100'	17.5	30.0	49.5	mA
	(DS3,DS2,DS1,DS0) = '1101'	18.8	32.2	53.0	mA
	(DS3,DS2,DS1,DS0) = '1110'	20.0	34.3	56.6	mA
	(DS3,DS2,DS1,DS0) = '1111'	21.2	36.4	60.1	mA
I_{OH}	High Level Output Current @ $V_{OH}(\min)$				
	(DS3,DS2,DS1,DS0) = '0001'	3.4	6.3	10.9	mA
	(DS3,DS2,DS1,DS0) = '0001'	5.2	9.4	16.4	mA
	(DS3,DS2,DS1,DS0) = '0010'	6.9	12.5	21.9	mA
	(DS3,DS2,DS1,DS0) = '0011'	8.6	15.6	27.3	mA
	(DS3,DS2,DS1,DS0) = '0100'	10.3	18.8	32.8	mA
	(DS3,DS2,DS1,DS0) = '0101'	12.0	21.9	38.2	mA
	(DS3,DS2,DS1,DS0) = '0110'	13.8	25.0	43.6	mA
	(DS3,DS2,DS1,DS0) = '0111'	15.5	28.1	49.1	mA
	(DS3,DS2,DS1,DS0) = '1000'	17.2	31.2	54.5	mA
	(DS3,DS2,DS1,DS0) = '1001'	18.9	34.3	60.0	mA
	(DS3,DS2,DS1,DS0) = '1010'	20.6	37.4	65.4	mA
	(DS3,DS2,DS1,DS0) = '1011'	22.3	40.6	70.9	mA
	(DS3,DS2,DS1,DS0) = '1100'	24.1	43.7	76.3	mA
	(DS3,DS2,DS1,DS0) = '1101'	25.8	46.8	81.7	mA
	(DS3,DS2,DS1,DS0) = '1110'	27.5	49.9	87.2	mA
	(DS3,DS2,DS1,DS0) = '1111'	29.2	53.0	92.6	mA

Note: The default drive configuration of (DS3,DS2,DS1,DS0) is "0010"

4.4.3 DDR I/O DC electrical characteristics

The following tables list parameters that are common across all memory interface standards. Unless otherwise specified, all input specifications (both common and DDR standard specific) are measured at the host PHY input pins.

Subsequent sections list parameters for specific memory interface standards.

4.4.3.1 DDR Input I/O DC electrical characteristics

Table 4-20 Common DC Input Conditions

Parameter	Symbol	Unit	Minimum	Typical	Maximum	Notes
Vref input leakage current	IIZ	uA			50	1
BP_DAT* input leakage current	IIZ	uA			50	2
Input high voltage for BP_DAT*	VIH-DC	V	Vref + 0.020			3,4
Input low voltage for BP_DAT*	VIL-DC	V			Vref - 0.020	3,4
Differential input voltage for abs(DQS _t - DQS _c)	VID-DC	V	0.100			5

Notes:

1. Leakage is valid for Vref over the range $0 \leq V_{IN} \leq V_{DDQ}$, with Vref input function enabled. All pins not under test = VDDQ.
2. Leakage current is measured when the pin is configured to a high-impedance state with all on-die termination disabled. Leakage is valid for any input except for Vref over the range: $0 \leq V_{IN} \leq V_{DDQ}$. All pins not under test = VSS or VDDQ.
3. The DC values indicate the voltage levels at which the final logic state of the receiver is unambiguously defined.
4. After running offset cancellation training sequence
5. The differential voltage VID is the difference between the true and complement signals. Absolute value taken so that minimum spec applies to both high and low differential inputs.
6. In DQS, PAD leakage would be twice the maximum BP_DAT leakage, as that have two attenuators.
7. Receiver attenuator leakage path can be disabled during test in certain configurations to reduce PAD leakage to maximum of 10uA. Note: The customer must keep the BP_DAT within VDD level in that case.
8. BP_DAT refers to DQS_{*}, DMI_{*} and DQ_{*}.

Table 4-21 Common DC Input Conditions for Flyover Bypass

Parameter	Symbol	Unit	Minimum	Typical	Maximum
Input high voltage	VIH_DC	V	0.7 * VDDQ		
Input low voltage	VIL_DC	V			0.3* VDDQ

Table 4-22 LPDDR4 DC Input Conditions

Parameter	Symbol	Unit	Mini.	Typical	Max.	Notes
Reference voltage	Vref	V		Variable		1
DC reference voltage error	VrefDC-err	VDDQ	-0.75%		0.75%	2
On-die termination (ODT) programmable resistances	RTT	ohm		open, 120, 60, 40		3, 4
ODT mid-level (0.25*VDDQ) large- signal resistance	RODT-M/R TT		0.8		1.1	5
ODT high-level (0.5*VDDQ)	RODT-H/R		0.9		1.1	5

large- signal resistance	TT					
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Notes:

1. BP_VREF must be set according to the DIMM loading and termination during reads. Because termination at the DRAMs is configurable, there is no fixed setting. The Vref value is dependent on driver impedance Ron and system effective ODT impedance Rtt. The Vref value for specific combination of Ron and Rodt can be calculated from the following equation:

$$V_{ref} = \frac{V_{DDQ}}{2} \left(\frac{R_{tt}}{R_{on} + R_{tt}} \right)$$

2. Externally supplied Vref is not recommended. Internal Vref generation through local Vref generation at each receiver is preferred.
3. The ODT level is programmable based on DIMM and DRAM configuration.
4. For LPDDR4, ODT is a pull-down to VSS.
5. ODT large-signal resistance is defined for an applied input voltage VIN and a measured input current of IIN as

$$R_{ODT} = \frac{V_{IN}}{I_{IN}}$$

4.4.3.2 DDR Output I/O DC electrical characteristics

Table 4-23 Common DC Output Parameters

Parameter	Symbol	Unit	Minimum	Typical	Maximum	Notes
Output driver pull-up impedance: DQ, DQS outputs	ROnPu	ohm		120, 60, 40		1
Output driver pull-down impedance: DQ, DQS outputs	ROnPd	ohm		120, 60, 40		1
Output driver pull-up impedance: address, command, CLK outputs	ROnPu	ohm		120, 60, 40		1
Output driver pull-down impedance: address, command, CLK outputs	ROnPd	ohm		120, 60, 40		1
Output driver pull-up impedance: CKE	ROnPu	ohm		24-40		3
Output driver pull-down impedance: CKE	ROnPd	ohm		24-40		3
Output driver pull-up impedance: RESET_N	ROnPu	ohm		24-40		4
Output driver pull-down impedance: RESET_N	ROnPd	ohm		24-40		4
Large-signal DQ, DQS pull-up variation at 0.25*VDDQ	ROnDQPu25 / ROn		0.9		1.3	2
Large-signal DQ, DQS pull-up variation at 0.5*VDDQ	ROnDQPu05 / ROn		0.9	1.0	1.1	2

Large-signal DQ, DQS pull-up variation at 0.8*VDDQ	ROnDQPu08 / ROn		0.8		1.05	2
Large-signal DQ, DQS pull-up variation at 1.1*VDDQ	ROnDQPu11 / ROn		0.75		1.05	2
Large-signal DQ, DQS pull-down variation at 0.25*VDDQ	ROnDQPd25 / ROn		0.80		1.1	2
Large-signal DQ, DQS pull-down variation at 0.5*VDDQ	ROnDQPd05 / ROn		0.90	1.0	1.1	2
Large-signal DQ, DQS pull-down variation at 0.8*VDDQ	ROnDQPd08 / ROn		0.90		1.35	2
Large-signal DQ, DQS pull-down variation at 1.1*VDDQ	ROnDQPd11 / ROn		0.95		1.75	2
Large-signal command, address pull-up variation at 0.25*VDDQ	ROnCAPu025 / ROn		0.9		1.35	2
Large-signal command, address pull-up variation at 0.5*VDDQ	ROnCAPu050 / ROn		0.9	1.0	1.1	2
Large-signal command, address pull-up variation at 0.75*VDDQ	ROnCAPu075 / ROn		0.8		1.1	2
Large-signal command, address pull-down variation at 0.25*VDDQ	ROnCAPd025 / ROn		0.8		1.1	2
Large-signal command, address pull-down variation at 0.5*VDDQ	ROnCAPd050 / ROn		0.9	1.0	1.1	2
Large-signal command, address pull-down variation at 0.75*VDDQ	ROnCAPd075 / ROn		0.9		1.35	2
Mismatch between pull-up and pull-down driver		%	-10		10	

Notes:

1. Calibrated at VDDQ / 2.
2. Large-signal output resistance is defined for an applied voltage VOUT and a measured current of IOU_T as

$$R_{ODT} = \frac{V_{DDQ} - V_{OUT}}{I_{OUT}}$$

3. For CKE the driver is always set to maximum strength and impedance value is process dependent
4. For RESET_N the driver is in maximum strength and impedance value is process dependent

4.5 I/O AC parameters

This section includes the DC parameters of the following I/O types:

- General Purpose I/O (GPIO) AC parameters

- SDIO AC parameters
- DDR I/O AC parameters

The GPIO and DDR I/O load circuit and output transition time waveforms are shown in Figure 4-3 and Figure 4-4.

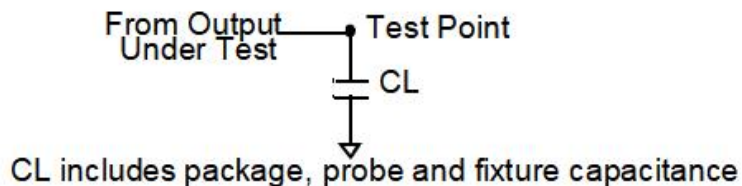


Figure 4-3 Load circuit for output

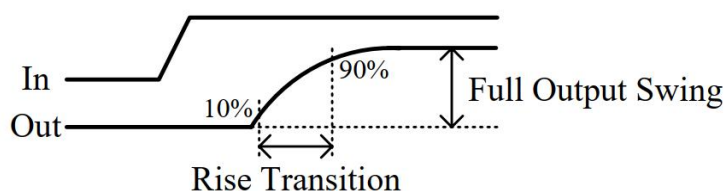


Figure 4-4 Output transition time waveform

4.5.1 General purpose I/O AC parameters

This section presents the I/O AC parameters for GPIO in different modes. Note that the fast or slow I/O behavior is determined by the appropriate control bits in the IOMUXC control registers.

Table 4-24 Maximum frequency of operation

NO.	Frequency (MHz)	Loading (pF)	IOMUXC Config (DS3,DS2,DS1,DS0)
1	25	19	0000
2	25	28	0001
3	25	38	0001
4	100	17	0010
5	125	19	0100
6	133	27	1000
7	163	14	0100
8	165	19	0110

Note:

- Process : tsmc 16nm
- Corner : SS
- Voltage : VDDPST=1.62V VDD=0.72V
- Temperature : 125°C

4.5.2 SDIO AC parameters

This section presents the I/O AC parameters for SDIO in different modes. Note that the fast or slow I/O behavior is determined by the appropriate control bits in the IOMUXC control registers.

Table 4-25 Maximum frequency of operation

NO.	Frequency (MHz)	Loading (pF)	IOMUXC Config (DS3,DS2,DS1,DS0)
1	100	17	1011
2	50	17	0100

Note:

- Process : tsmc 16nm
- Corner: : SS
- Voltage : VDDPST33=2.7V VDDPST18=1.35V VDD=0.72V
- Temperature : 125°C

Table 4-26 Maximum frequency of operation

NO.	Frequency (MHz)	Loading (pF)	IOMUXC Config (DS3,DS2,DS1,DS0)
1	100	17	0110
2	50	17	0011

Note:

- Process : tsmc 16nm
- Corner: : SS
- Voltage : VDDPST33=1.62V VDDPST18=1.62V VDD=0.72V
- Temperature : 125°C

4.5.3 DDR I/O AC parameters

The following tables list parameters that are common across all memory interface standards. Unless otherwise specified, all input specifications (both common and DDR standard specific) are measured at the host PHY input pins.

Subsequent sections list parameters for specific memory interface standards.

4.5.3.1 DDR Input I/O AC electrical characteristics**Table 4-27 Common AC Input Conditions**

Parameter	Symbol	Unit	Minimum	Typical	Maximum	Notes
Input capacitance DQ	C _{pad} DQ	pF	1.71	1.81	1.93	1,2
Input capacitance, address/cmd	C _{pad} AC	pF	1.63	1.71	1.74	1
Input capacitance DQS	C _{pad} DQDQS	pF	1.94	2.02	2.12	1,2

Notes:

1. This is specified at the top level of the DBYTE and ACx4 blocks. It does not include any customer-supplied routing between these blocks and the C4 bump.
2. Effective parallel capacitance when ODT of 60ohm is enabled.

Table 4-28 LPDDR4 AC Input Conditions

Parameter	Symbol	Unit	Minimum	Typical	Maximum	Notes
Reference voltage	VrefAC-err	VDDQ	-0.25%		0.25%	1
Low-speed input high	VIH-ACLS	V	Vref+0.070		VDDQ +	2

voltage for BP_DAT (logic 1)					0.15	
Low-speed input low voltage for BP_DAT (logic 0)	VIL-ACLS	V	-0.15		Vref - 0.070	2
High-speed input high voltage for BP_DAT (logic 1)	VIH-ACHS	V	Vref + 0.070		Vref + 0.440	3
High-speed input low voltage for BP_DAT (logic 0)	VIL-ACHS	V	Vref-0.440 or 0		Vref - 0.070	3
ODT mid-level (0.25*VDDQ) small-signal resistance	RODTss-M/RTT		0.9		1.1	4
ODT high-level (0.5*VDDQ) small-signal resistance	RODTss-H/RTT		0.95		1.75	4
Input crossing voltage for differential signals DQS*	VIX	mV	VDQSavg -75	VDQSavg	VDQSavg +75	5

Notes:

1. Peak-to-peak noise on the Vref input, normalized to VDDQ
2. This is the voltage at which the input receiver must be able to discriminate a high or a low input during a transient switching event. The wider signal swing allowance causes a timing penalty due to the larger amount of cap charging and discharging in the receiver.
3. This narrower signal swing range incurs no timing penalty. Minimum signal must be > VSS.
4. Small-signal resistance is defined, for measured input current IIN as a function of applied voltage VIN, as the first derivative of the V-I plot:

$$R_{ODTss} = \frac{dV_{IN}}{dI_{IN}}$$

5. VDQSavg is defined as the average of 0.5*(DQS_t + DQS_c).
6. BP_DAT refers to DQS_*, DMI_* and DQ_*.

4.5.3.2 DDR Output I/O AC electrical characteristics

Table 4-29 Common AC Output Parameters

Parameter	Symbol	Unit	Mini	Typical	Max	Notes
Small-signal DQ, DQS pull-up variation at 0.25*VDDQ	RONssDQPu25 / RON		0.95		5.0	1,2
Small-signal DQ, DQS pull-up variation at 0.5*VDDQ	RONssDQPu05 / RON		0.95		1.75	1
Small-signal DQ, DQS pull-up variation at 0.8*VDDQ	RONssDQPu08 / RON		0.85		1.05	1
Small-signal DQ, DQS pull-up variation at 1.1*VDDQ	RONssDQPu11 / RON		0.75		1.05	1
Small-signal DQ, DQS pull-down variation at 0.25*VDDQ	RONssDQPd25 / RON		0.9		1.1	1
Small-signal DQ, DQS pull-down variation at 0.5*VDDQ	RONssDQPd05 / RON		0.95		1.75	1
Small-signal DQ, DQS pull-down variation at 0.8*VDDQ	RONssDQPd08 / RON		0.95		8.0	1,2
Small-signal DQ, DQS pull-down	RONssDQPd11 /		1.0		30	1, 2

variation at 1.1*VDDQ	R _{On}					
Small-signal command, address pull-up variation at 0.25*VDDQ	R _{OnssCAPu025} / R _{On}		0.95		6.5	1
Small-signal command, address pull-up variation at 0.5*VDDQ	R _{OnssCAPu050} / R _{On}		0.95		2.0	1
Small-signal command, address pull-up variation at 0.75*VDDQ	R _{OnssCAPu075} / R _{On}		0.75		1.1	1
Small-signal command, address pull-down variation at 0.25*VDDQ	R _{OnssCAPd025} / R _{On}		0.75		1.1	1
Small-signal command, address pull-down variation at 0.5*VDDQ	R _{OnssCAPd050} / R _{On}		0.95		2.5	
Small-signal command, address pull-down variation at 0.75*VDDQ	R _{OnssCAPd075} / R _{On}		0.95		6.5	

Notes:

- Small-signal resistance is defined, for measured input current I_{IN} as a function of applied voltage V_{IN}, as the first derivative of the V-I plot:

$$R_{ODTss} = \frac{dV_{IN}}{dI_{IN}}$$

- These numbers are extremely large because the curvature of the transistor characteristics dominates the I-V curve at large voltages. However, since the current at the calibrated 0.5*VDDQ point sets the output level, the curvature will have minimal effect on the signaling.

Table 4-30 Output Timing Characteristics

Parameter	Symbol	Unit	Minimum	Typical	Maximum	Notes
Skew across address lanes in an Address block	tADDRskew	ps			0.25*(Pclk period)	1
M_DQS high pulse width	tDQSH	ps	0.46*tck		0.54*tck	
M_DQS low pulse width	tDQSL	ps	0.46*tck		0.54*tck	

Notes:

- This timing parameter applies to all address, command, and control signals.

Table 4-31 Common DC Output Conditions for Flyover Bypass1

Parameter	Symbol	Unit	Minimum	Typical	Maximum
Output high voltage	VOH_DC	V	0.8 * VDDQ		
Output low voltage	VOL_DC	V			0.2 * VDDQ

Notes:

VOH_DC and VOL_DC for Flyover Bypass mode limit is without termination.

Table 4-32 Common DC and AC Output Conditions for Reset_n¹

Parameter	Symbol	Unit	Minimum	Typical	Maximum
Output high voltage	VOH_DC	V	0.7 * VDDQ		VDDQ
Output low voltage	VOL_DC	V	VSS		0.3* VDDQ
Output high voltage	VOH_AC	V	0.8 * VDDQ		VDDQ
Output low voltage	VOL_AC	V	VSS		0.2* VDDQ

Notes:

Reset_n drives full CMOS level without termination.

4.6 System modules timing

4.6.1 Reset timings parameters

The following figure shows the reset timing and table lists the timing parameters.

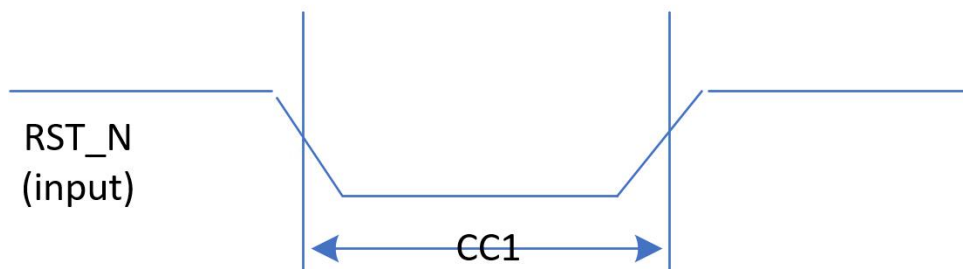


Figure 4-5 Reset timing diagram

Table 4-33 Table Reset timing parameter

ID	Parameter	Min	Max	Unit
CC1	Duration of RST_N to be qualified as 0.	60	-	ns

4.6.2 WDOG Reset timing parameters

The following figure shows the WDOG reset timing and table lists the timing parameters.

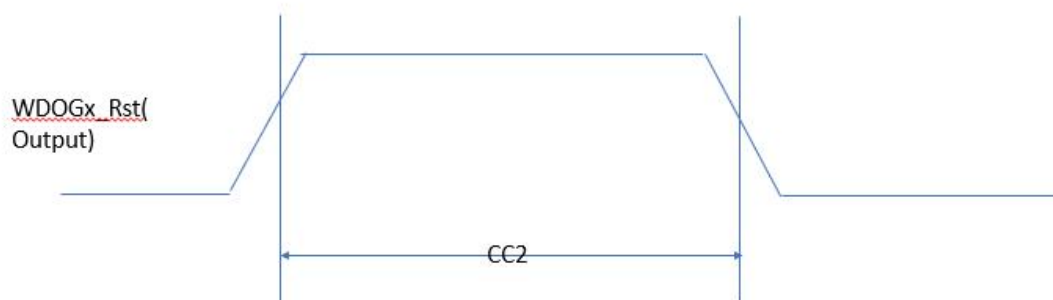


Figure 4-6 WDOG timing diagram

Table 4-34 Table Reset timing parameter

ID	Parameter	Min	Max	Unit
CC2	Duration of WDOGx_Rst Assertion	4	256	XTAL cycle

4.7 External peripheral interface parameters

4.7.1 JTAG timing parameters

The JTAG (IEEE Standard 1149.1-1990 Standard-Test-Access Port and Boundary Scan Architecture) interface is used for BSDL testing and emulation of the device. The trstn pin only needs to be released when it is necessary to use a JTAG controller to debug the device or exercise the device's boundary scan functionality. For maximum reliability, the device includes an Internal Pulldown (IPD) on the trstn pin to ensure that trstn is always asserted upon power up and the device's internal emulation logic is always properly initialized. JTAG controllers from Texas Instruments actively drive trstn high. However, some third-party JTAG controllers may not drive trstn high but expect the use of a Pullup resistor on trstn. When using this type of JTAG

controller, assert trstn to initialize the device after powerup and externally drive trstn high before attempting any emulation or boundary-scan operations.

4.7.1.1 JTAG Electrical Data/Timing

Table 4-35 Timing Requirements for IEEE 1149.1 JTAG

NO.	PARAMETER	DESCRIPTION	MIN	MAX	UNIT
1	$t_c(\text{TCK})$	Cycle time, TCK	62.29		ns
1H	$t_w(\text{TCKH})$	Pulse duration, TCK high (40% of t_c)	24.92		ns
1L	$t_w(\text{TCKL})$	Pulse duration, TCK low (40% of t_c)	24.92		ns
3	$t_{su}(\text{TDI-TCK})$	Input setup time, TDI valid to TCK high	6.23		ns
	$t_{su}(\text{TMS-TCK})$	Input setup time, TMS valid to TCK high	6.23		ns
4	$t_h(\text{TCK-TDI})$	Input hold time, TDI valid from TCK high	31.15		ns
	$t_h(\text{TCK-TMS})$	Input hold time, TMS valid from TCK high	31.15		ns

Table 4-36 Switching Characteristics Over Recommended Operating Conditions for IEEE 1149.1 JTAG

NO.	PARAMETER	DESCRIPTION	MIN	MAX	UNIT
2	$t_d(\text{TCKL-TDOV})$	Delay time, TCK low to TDO valid	0	30.5	ns

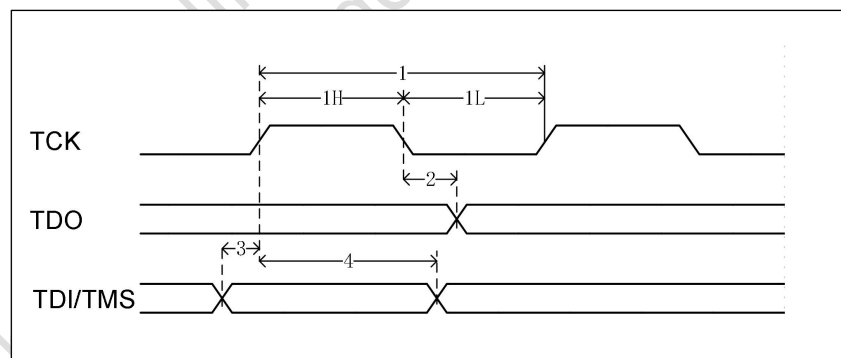


Figure 4-7 JTAG Timing

4.7.2 UART timing parameters

Because the serial communication between A1000L uart and a selected device is asynchronous, additional bits (start and stop) are added to the serial data to indicate the beginning and end. Utilizing these bits allows two devices to be synchronized. This structure of serial data-accompanied by start and stop bits-is referred to as a character, as shown in Figure 4-8.

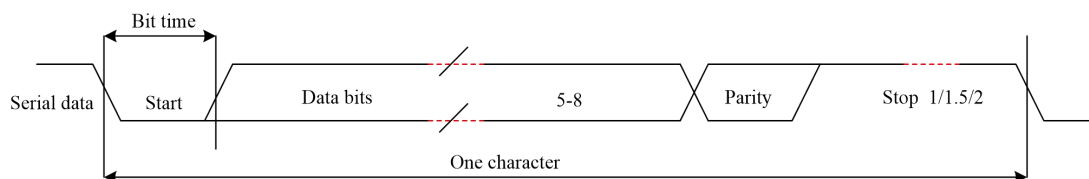


Figure 4-8 Serial data format

Figure 4-9 depicts the transmit timing of UART in the RS-232 Serial mode, with 8 data bit/1 stop bit format. Table 4-38 lists the UART RS-232 Serial mode transmit timing characteristics.

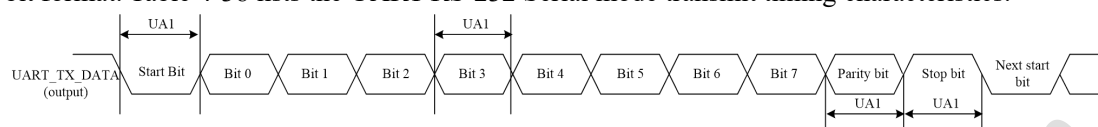


Figure 4-9 UART RS-232 Serial mode transmit timing

Table 4-37 Serial mode transmit timing parameters

ID	Parameter	Symbol	Min	Max	Unit
UA1	Transmit Bit time	tTbit	$1/\text{Fbaud_rate}(1) - \text{Tref_clk}(2)$	$1/\text{Fbaud_rate} + \text{Tref_clk}$	-

(1) . Fbaud_rate: Baud rate frequency. The maximum baud rate the UART can support is $(\text{ipg_perclk frequency})/16$.

(2) . Tref_clk: The period of UART reference clock ref_clk (ipg_perclk after RFDIV divider).

Figure 4-10 depicts the RS-232 Serial mode receive timing with 8 data bit/1 stop bit format.

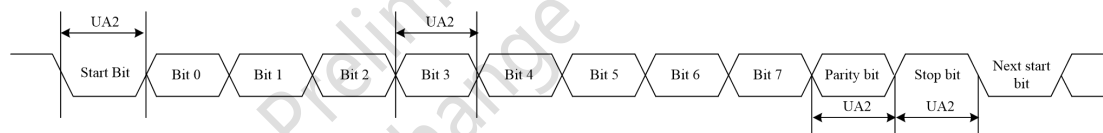


Figure 4-10 UART RS-232 Serial mode receive timing

Table 4-38 RS-232 Serial mode receive timing parameters

ID	Parameter	Symbol	Min	Max	Unit
UA2	Receive Bit time	tRbit(1)	$1/\text{Fbaud_rate}(2) - 1/(16 \times \text{Fbaud_rate})$	$1/\text{Fbaud_rate} + 1/(16 \times \text{Fbaud_rate})$	-

(1).The UART receiver can tolerate $1/(16 \times \text{Fbaud_rate})$ tolerance in each bit. But accumulation tolerance in one frame must not exceed $3/(16 \times \text{Fbaud_rate})$.

(2).Fbaud_rate: Baud rate frequency. The maximum baud rate the UART can support is 115200

4.7.3 I2C timing parameters

The Inter-Integrated Circuit (I2C) provides functionality of a standard I2C master and slave. The I2C is designed to be compatible with the I2C Bus Specification, version 3.0, dated April 2014.

The A1000L project I2C support three speeds, as following.

- (1). Standard mode (0 to 100 Kb/s) .
- (2). Fast mode ($\leq 400 \text{ Kb/s}$) or fast mode plus ($\leq 1000 \text{ kb/s}$)
- (3). high-speed mode ($\leq 3.4 \text{ Mb/s}$)

4.7.4 SPI timing parameters

The Inter-Integrated Circuit(SPI) provides functionality of a standard SPI transmitter and receive. The SPI is designed to be compatible with the SPI Bus Specification, Based on Motorola.

The timing as the following diagram for spi.

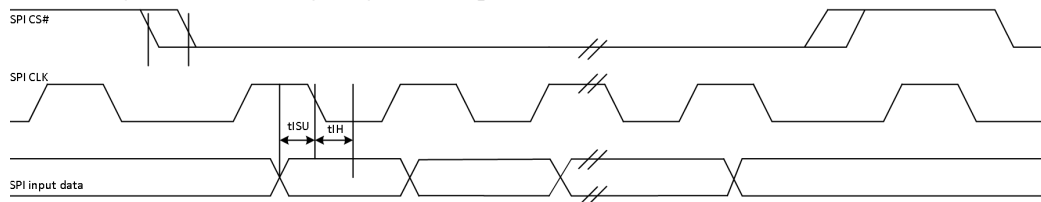


Figure 4-11 SPI input timing

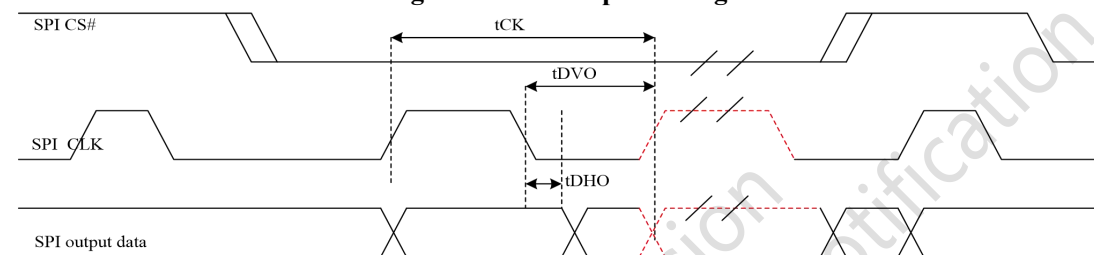


Figure 4-12 SPI output timing

The timing parameter for SPI as the following table.

Table 4-39 SPI master timing parameter

	Value		Unit	Description
	min	max		
Clock period T/tCK	40		ns	
Clock Duty cycle ratio	45/55	55/45	%	
MISO signal/input signal				
tISU	31	-	ns	Input setup time
tIH	1	-	ns	Input hold time
MOSI, CS signal/output signal				
tDVO	-	25	ns	Output data valid time
tDHO	15	-	ns	Output data hold time

4.7.5 I2S timing parameters

A1000L component interfaces to the I2S bus, which is protected by patents held by Philips (NXP). And The bus consists of a serial data line (sd), a word selects line (ws), and a serial clock (sclk). The serial data line is time multiplexed to allow the transfer of two data streams (such as, left and right stereo data). The I2S bus stereo frame format as the following diagram.

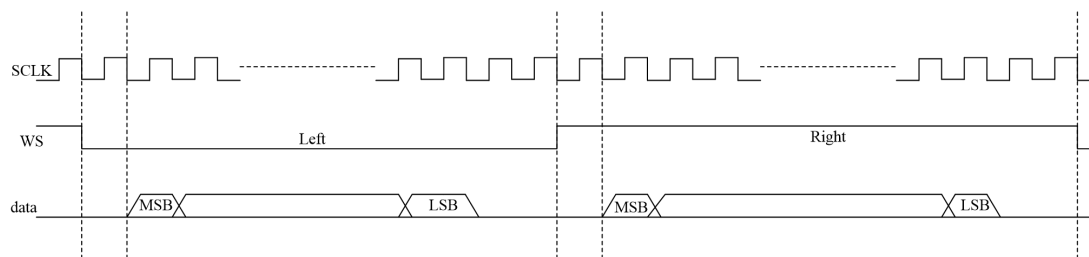


Figure 4-13 I2S bus stereo frame format

The bus for transmitter timing as the Figure 4-14.

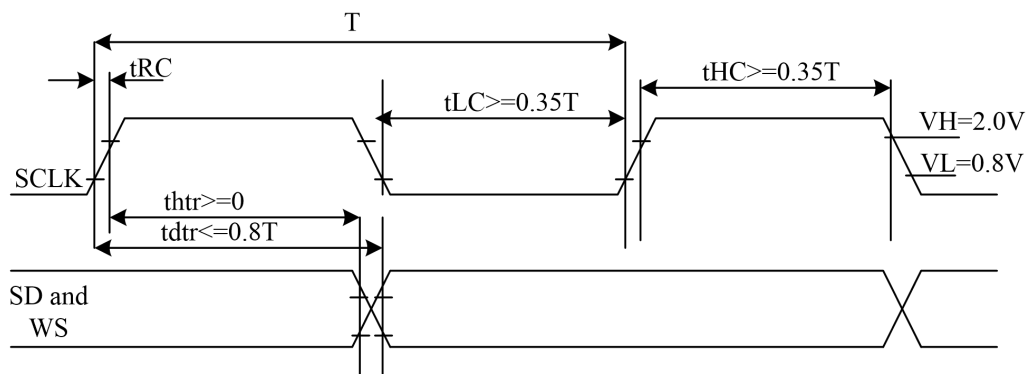


Figure 4-14 Timing for I2S transmitter

The bus for receiver timing as the Figure 4-15.

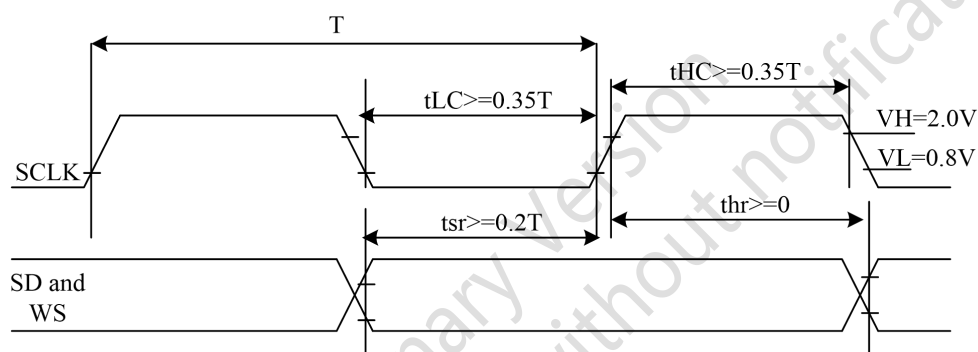


Figure 4-15 Timing for I2S receiver

Note that the times given in both Figure 4-14 and Figure 4-15 are defined by the transmitter speed.

The specification of the receiver must be able to match the performance of the transmitter. The following table example with data rate of 2.5MHz($\pm 10\%$) for i2s master and slave.

Table 4-40 Master transmitter with data rate of 6.144MHz ($\pm 10\%$) (all values in ns)

	MIN	TYP	MAX	CONDITION
clock period T	146.484	162.76	179.036	$T_{tr} = 146.484$
clock HIGH t _{HC}	65.104			$\min > 0.35T = 56.966$ (at typical data rate)
clock LOW t _{LC}	65.104			$\min > 0.35T = 56.966$ (at typical data rate)
delay t _{dtr}			122.07	$\max < 0.80T = 130.208$ (at typical data rate)
hold time t _{thr}	40.69			$\min > 0$
clock rise-time t _{RC}			24.414	$\max > 0.15T_{tr} = 21.976$ (only relevant in slave mode)

Table 4-41 Slave receive with data rate of 6.144MHz ($\pm 10\%$) (all values in ns)

	MIN	TYP	MAX	CONDITION
clock period T	146.484	162.76	179.036	$T_{tr} = 146.484$
clock HIGH t _{HC}	44.759			$\min < 0.35T = 56.966$

clock LOW tLC	44.759			$\min < 0.35T = 56.966$
set-up time tsr	21.414			$\min < 0.2T = 32.552$
hold time thtr	0			$\min < 0$

And the timing parameter for i2s transmitter and receives as the following table.

Table 4-42 Timing parameter for i2s transmitter and receives

	transmitter				receiver				Notes
	low limit		upper limit		low limit		upper limit		
	min	max	min	max	min	max	min	max	
Clock period T	Ttr				Tr				1
MASTER mode: clock generated by transmitter or receiver									
HIGH tHC	0.35Ttr				0.35Ttr				2a
LOW tLC	0.35Ttr				0.35Ttr				2a
SLAVE MODE: clock accepted by transmitter or receiver.									
HIGH tHC		0.35Ttr				0.35Ttr			2b
LOW tLC		0.35Ttr				0.35Ttr			2b
rise-time tRC			0.15Ttr						3
TRANSMITTER:									
delay tdr				0.8T					4
Hold time thtr	0								3
RECEIVER:									
set-up time tsr						0.2Tr			5
hold time thr						0			5

4.7.6 PWM timing parameters

The PWM can be programmed to select one of three clock signals as its source frequency. The selected clock signal is passed through a prescaler before being input to the counter. The output is available at the pulse-width modulator output (PWM_OUT) external pin.

Figure 4-16 depicts the timing of the PWM, and Table 4-44 lists the PWM timing parameters.

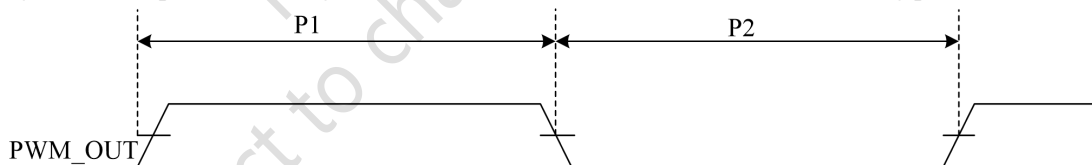


Figure 4-16 PWM timing

Table 4-43 PWM output timing parameters

ID	Parameter	Min	Max	Unit
	PWM output clock frequency		100MHz	MHz
P1	PWM output pulse width high	5	-	ns
P2	PWM output pulse width low	5	-	ns

4.7.7 OSPI timing parameters

A1000L need the OSPI support SDR mode, the following describe the SDR mode timing parameter. Figure 4-17 is the SDR mode input timing and Figure 4-18 is the SDR mode output timing.

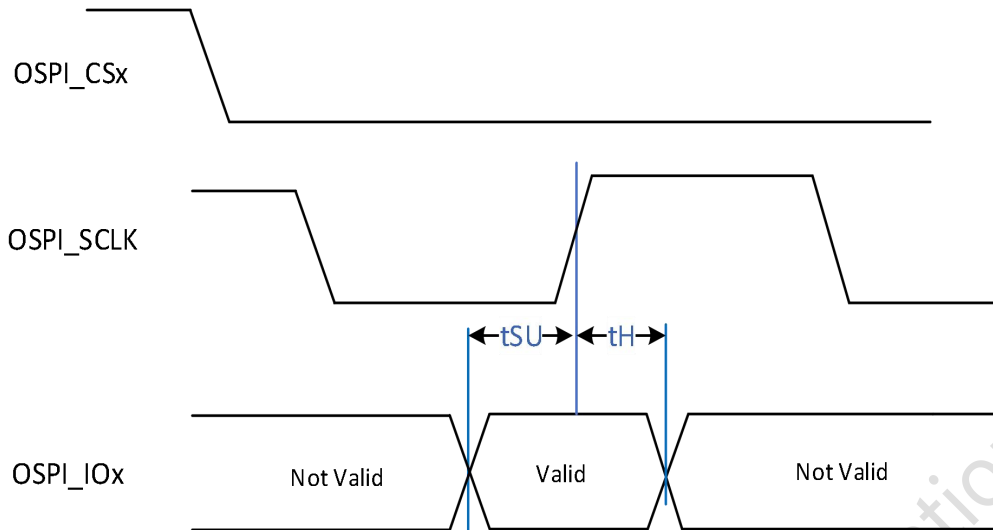


Figure 4-17 SDR mode input timing

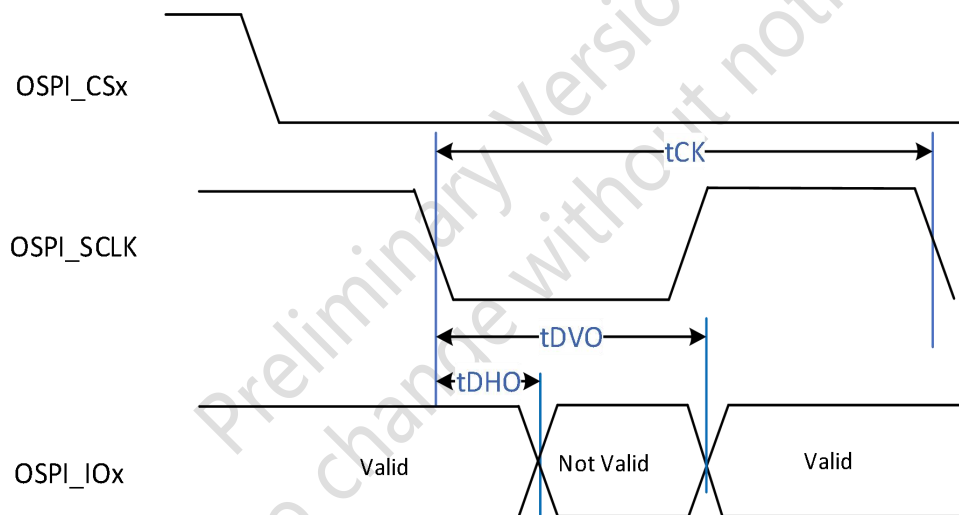


Figure 4-18 SDR mode output timing

The OSPI timing parameter as the following table.

Table 4-44 OSPI timing parameter

Parameter	Symbol	Min.	Max.	Unit	Remark
Clock					
Clock frequency	tCK	0	100	MHz	
Duty cycle ratio	-	45/55	55/45	%	
Inputs data (referenced to OSPI clock)					
Input setup time	tSU	4.6	-	ns	
Input hold time	tH	1	-	ns	
Outputs data (referenced to OSPI clock)					
Output data valid time	tDVO	-	8.3	ns	
Output data hold time	tDHO	2.7	-	ns	

Note:

Timing requirement: the delay difference between data and clock should not exceed 200ps

4.7.8 CAN FD timing parameters

CAN FD is a pure asynchronous half duplex communication bus, and the protocol defines two of bit rates, one bit rates is a nominal bit rates, and the other is data bit rates. Nominal bit rates is a long time bit rate, and the time of data bit rates is less than nominal bit rates.

CAN FD protocol defines a sample point to sample data, and the detail of the sample point please reference the CAN FD protocol which release by BOSH.

The CAN FD main support 125kbps/250kbps/500kbps/1Mbps/2Mbps/4Mbps/5Mbps/8Mbps/10Mbps bit rate in A1000L. The sample clock frequency is 200MHz, and the timing is shown in follow figure.

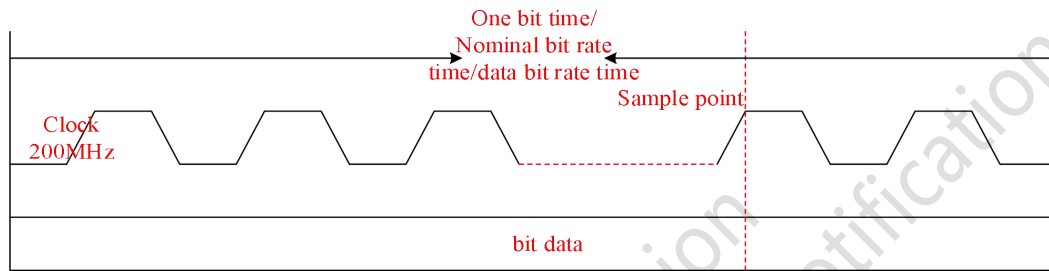


Figure 4-19 CANFD bus sample timing

4.7.9 SDEMMC timing parameters

4.7.9.1 SD controller timing

A1000L supports five modes: DS, HS, SDR12, SDR25, SDR50. The SD controller input timing diagram as following.

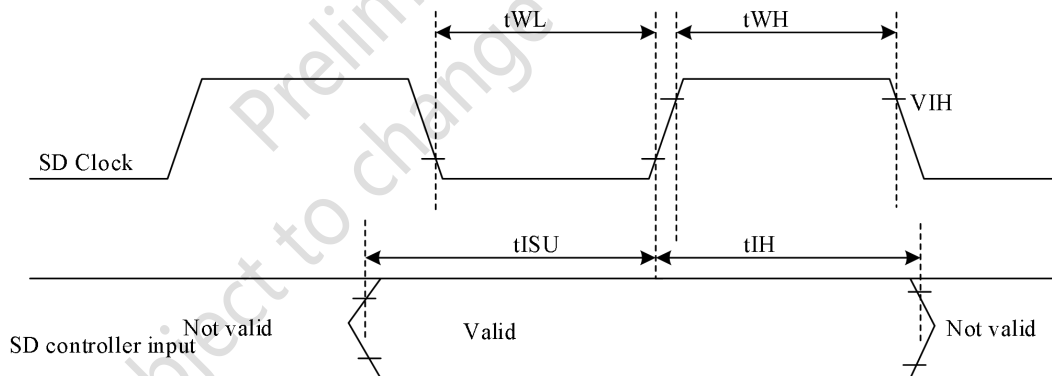


Figure 4-20 SD input timing

The SD controller output timing diagram as following.

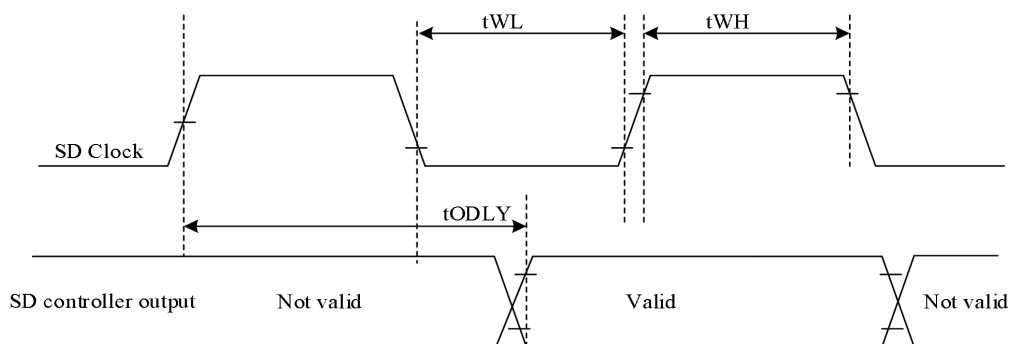


Figure 4-21 SD output timing

The SD timing parameter as the following table.

Table 4-45 DS/HS/SDR12/SDR25/SDR50 mode timing parameter

Parameter	Symbol	Min.	Max.	Unit	Remark
Max clock frequency			100	Mhz	SDR50
Input setup time	tISU	2.2	-	ns	
Input hold time	tIH	1.1	-	ns	
Clock duty cycle ratio	tWL/tWH	45/55	55/45	%	
Output delay	tODLY	1.2	6.6	ns	

Note:

Timing requirement: the delay difference between data and clock should not exceed 200ps

4.7.9.2 EMMC timing parameters

The eMMC defines two bus speed modes. Following table summarizes the various modes.

Table 4-46 EMMC bus speed modes summarizes

Mode name	Data rate	I/O voltage	Bus width	Frequency	Max data transfer
Backwards Compatibility with legacy MMC card	Single	1.8	8	0-25MHz	25MB/s
High speed SDR	Single	1.8	8	0-50MHz	50MB/s

Figure 4-22 shows the data input/output timing and note that data must always be sampled on the rising edge of the clock.

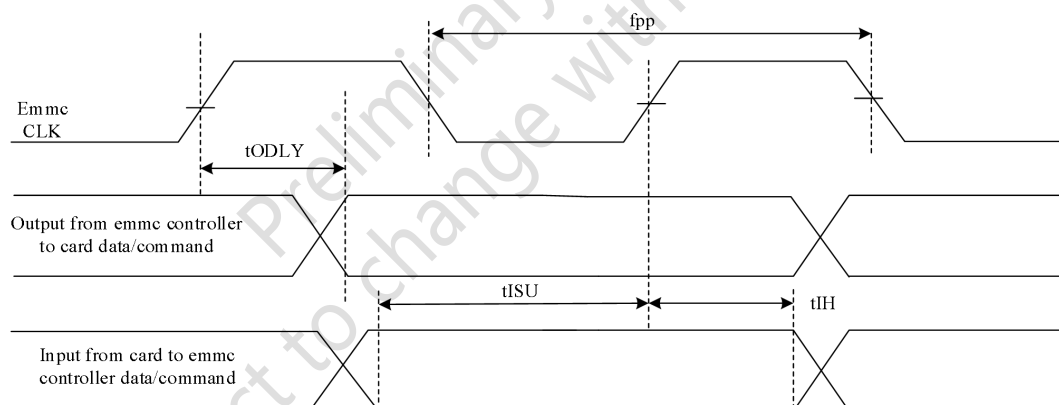


Figure 4-22 Data input/output timing

Table 4-47 Timing parameters of EMMC

Parameter	Symbol	Min.	Max.	Unit	Remark
Max clock frequency	fpp	-	50	Mhz	
Input setup time	tISU	5.7	-	ns	
Input hold time	tIH	1.9	-	ns	
Clock duty cycle ratio		45/55	55/45	%	
Output delay	tODLY	3.6	16.4	ns	

Note:

- 1) In identification mode (SD), card clock must be lower than 400 kHz, voltage ranges from 2.7 to 3.6 V.
- 2) To satisfy timing requirement, the delay difference between clock and cmd/data must not exceed 200ps.

4.7.10 ERR_RPT timing parameters

NA.

4.7.11 GMAC timing parameters

4.7.11.1 GMAC signals map

Table 4-48 GMAC signals map

Ball Name	Description	Mode	Direction	Notes
ETH0_TXD0	GMII/RGMII/MII.TXD0	GMII/RGMII/MII	O	GMII/RGMII/MII transmit data
ETH0_TXD1	GMII/RGMII/MII.TXD1	GMII/RGMII/MII	O	GMII/RGMII/MII transmit data
ETH0_TXD2	GMII/RGMII/MII.TXD2	GMII/RGMII/MII	O	GMII/RGMII/MII transmit data
ETH0_TXD3	GMII/RGMII/MII.TXD3	GMII/RGMII/MII	O	GMII/RGMII/MII transmit data
ETH0_TXD4	GMII.TXD4	GMII	O	GMII transmit data
ETH0_TXD5	GMII.TXD5	GMII	O	GMII transmit data
ETH0_TXD6	GMII.TXD6	GMII	O	GMII transmit data
ETH0_TXD7	GMII.TXD7	GMII	O	GMII transmit data
ETH0_TXER	GMII/MII.TXER	GMII/MII	O	GMII or MII transmit error
ETH0_TXCTRL	GMII/MII.TXEN, RGMII.TX_CTL	GMII/RGMII/MII	O	GMII or MII transmit enable, RGMII transmit control
ETH0_GTXCLK	GMII/RGMII.GTXCLK	GMII/RGMII	O	GMII or RGMII transmit clock
ETH0_TXCLK	MII.TXCLK	MII	I	MII transmit clock
ETH0_RXD0	GMII/RGMII/MII.RXD0	GMII/RGMII/MII	I	GMII/RGMII/MII receive data
ETH0_RXD1	GMII/RGMII/MII.RXD1	GMII/RGMII/MII	I	GMII/RGMII/MII receive data
ETH0_RXD2	GMII/RGMII/MII.RXD2	GMII/RGMII/MII	I	GMII/RGMII/MII receive data
ETH0_RXD3	GMII/RGMII/MII.RXD3	GMII/RGMII/MII	I	GMII/RGMII/MII receive data
ETH0_RXD4	GMII.RXD4	GMII	I	GMII receive data
ETH0_RXD5	GMII.RXD5	GMII	I	GMII receive data
ETH0_RXD6	GMII.RXD6	GMII	I	GMII receive data
ETH0_RXD7	GMII.RXD7	GMII	I	GMII receive data
ETH0_RXER	GMII/MII.RXER	GMII/MII	I	GMII or MII receive error
ETH0_RXCTRL	GMII/MII.RXDV, RGMII.RX_CTL	GMII/RGMII/MII	I	GMII or MII receive valid, RGMII receive control
ETH0_RXCLK	GMII/RGMII/MII.RXCLK	GMII/RGMII/MII	I	GMII/RGMII/MII receive clock
ETH0_MDC	ETH0.MDC	GMII/RGMII/MII	O	
ETH0_MDIO	ETH0.MDIO	GMII/RGMII/MII	I/O	
ETH0_INTR	ETH0.INTR	GMII/RGMII/MII	I	ETH0 interrupt input,

		MII		active high is valid
ETH1_TXD0	RGMII/MII.TXD0	RGMII/MII	O	RGMII/MII transmit data
ETH1_TXD1	RGMII/MII.TXD1	RGMII/MII	O	RGMII/MII transmit data
ETH1_TXD2	RGMII/MII.TXD2	RGMII/MII	O	RGMII/MII transmit data
ETH1_TXD3	RGMII/MII.TXD3	RGMII/MII	O	RGMII/MII transmit data
ETH1_TXER	MII.TXER	MII	O	MII transmit error
ETH1_TXCTRL	MII.TXEN, RGMII.TX_CTL	RGMII/MII	O	MII transmit enable, RGMII transmit control
ETH1_GTXCLK	RGMII.GTXCLK	RGMII	O	RGMII transmit clock
ETH1_TXCLK	MII.TXCLK	MII	I	MII transmit clock
ETH1_RXD0	RGMII/MII.RXD0	RGMII/MII	I	RGMII/MII receive data
ETH1_RXD1	RGMII/MII.RXD1	RGMII/MII	I	RGMII/MII receive data
ETH1_RXD2	RGMII/MII.RXD2	RGMII/MII	I	RGMII/MII receive data
ETH1_RXD3	RGMII/MII.RXD3	RGMII/MII	I	RGMII/MII receive data
ETH1_RXER	MII.RXER	RGMII/MII	I	MII receive error
ETH1_RXCTRL	MII.RXDV, RGMII.RX_CTL	RGMII/MII	I	MII receive valid, RGMII receive control
ETH1_RXCLK	RGMII/MII.RXCLK	RGMII/MII	I	RGMII/MII receive clock
ETH1_MDC	ETH1.MDC	RGMII/MII	I/O	
ETH1_MDIO	ETH1.MDIO	RGMII/MII	I	
ETH1_INTR	ETH1.INTR	RGMII/MII	I	ETH1 interrupt input, active high is valid
TS_TRIG_IN00	ETH0.TRIG_IN0	PTP	I	ETH0 Timestamp Trigger input. When this signal goes high (rising edge), a snapshot of the System Time is taken and stored in the auxiliary timestamp FIFO.
TS_TRIG_IN01	ETH0.TRIG_IN1	PTP	I	ETH0 Timestamp Trigger input. When this signal goes high (rising edge), a snapshot of the System Time is taken and stored in the auxiliary timestamp FIFO.
TS_TRIG_IN10	ETH1.TRIG_IN0	PTP	I	ETH1 Timestamp Trigger input. When this signal goes high (rising edge), a snapshot of the System Time is taken and stored in the auxiliary timestamp FIFO.
TS_TRIG_IN11	ETH1.TRIG_IN1	PTP	I	ETH1 Timestamp Trigger input. When this signal goes high (rising edge), a snapshot of the System Time is taken and stored in the auxiliary timestamp FIFO.
PTP_PPS0	ETH0.PPS	PTP	O	ETH0 output Pulse Per Second

PTP_PPS1	ETH1.PPS	PTP	O	ETH1 output Pulse Per Second
PTP_CLK	ETH.PTP_CLK	PTP	I	PTP Reference Clock. This reference clock, provided by an external oscillator or a timing source, can be used for the timestamp logic. The timestamp logic can select clock source from PTP_CLK, XTAL_IN, ETH0_RXCLK, ETH1_RXCLK

4.7.11.2 RGMII timing parameters

Table 4-49 RGMII signal switching specifications

Symbol	Description	Min.	Max.	Unit
T_{skewT}^1	Data to Clock output Skew (at Transmitter)	-800	800	ps
T_{skewR}^1	Data to Clock input Skew (at Receiver)	1	3	ns
T_{cyc}^2	Clock Cycle Duration	7.2	8.8	ns
Duty _G ³	Duty Cycle for Gigabit	45	55	%
Duty _T ³	Duty Cycle for 10/100T	40	60	%
Tr / Tf	Rise / Fall Time (20-80%)		.75	ns

Note:

1: For all versions of RGMII prior to 2.0; This implies that PC board design will require clocks to be routed such that an additional trace delay of greater than 1.5ns and less than 2.0ns will be added to the associated clock signal. For 10/100 the Max value is unspecified. There are +/-0.2ns skew margin on board to meet RGMII PHY standard timing.

2: For 10Mbps and 100Mbps, T_{cyc} will scale to 400ns+/-40ns and 40ns+/-4ns respectively.

3: Duty cycle may be stretched/shrunk during speed changes or while transitioning to a received packet's clock domain as long as minimum duty cycle is not violated and stretching occurs for no more than three T_{cyc} of the lowest speed transitioned between.

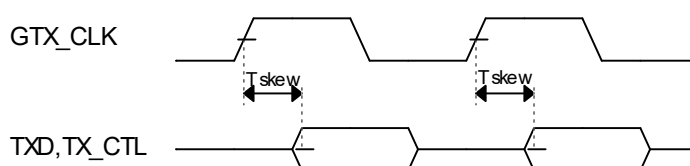


Figure 4-23 RGMII transmit signal timing diagram

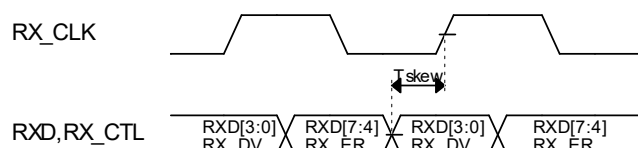


Figure 4-24 RGMII receive signal timing diagram

4.7.11.3 GMII timing parameters

Table 4-50 GMII signal switching specifications

Symbol	Description	Min.	Max.	Unit
f_{Freq}^1	GTX_CLK, RX_CLK Frequency	125 –100 ppm	125 +100 ppm	MHz
T_{skew}^1	Data to Clock output Skew (at Transmitter)	-1	1	ns
t_{SETUP}^1	RXD, RX_DV, RX_ER Setup to RX_CLK ring edge	1.0	-	ns
t_{HOLD}^2	RXD, RX_DV, RX_ER hold from RX_CLK ring edge	1.0	-	ns

Note:

1. For meeting IEEE802.3 GMII timing, add 4ns delay for GTX_CLK more than the delay of TXD, TX_ER, TXDV on the board, or PHY use falling edge of the clock to sample the data.
2. For meeting IEEE802.3 GMII timing, board or PHY add 1ns delay for RXD, RX_DV, RX_ER than the delay of RX_CLK.



Figure 4-25 GMII transmit signal timing diagram

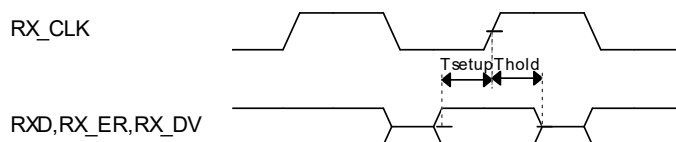


Figure 4-26 GMII receive signal timing diagram

Note: Only GMAC0 can support GMII.

4.7.11.4 MII timing parameters

Table 4-51 MII signal switching specifications

Symbol	Description	Min.	Max.	Unit
f_{Freq}	TX_CLK, RX_CLK Frequency	25 –100 ppm	25 +100 ppm	MHz
T_{skew}	Clock to data output Skew	0	25	ns
t_{SETUP}	RXD, RX_DV, RX_ER Setup to RX_CLK ring edge	10	-	ns
t_{HOLD}	RXD, RX_DV, RX_ER hold from RX_CLK ring edge	10	-	ns

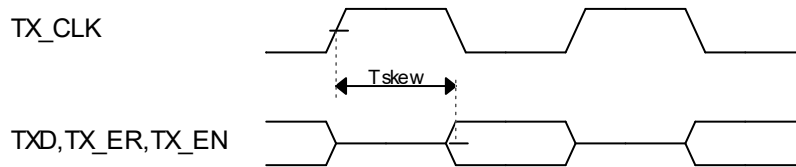


Figure 4-27 MII transmit signal timing diagram

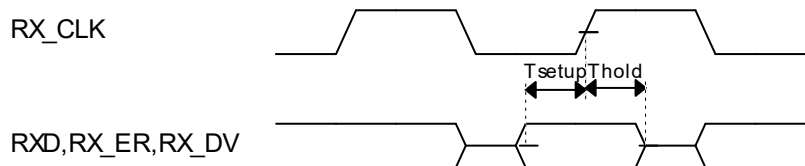


Figure 4-28 MII receive signal timing diagram

4.7.11.5 MDIO timing parameters

Table 4-52 MDIO signal switching specifications

Symbol	Description	Min.	Max.	Unit
f_{FREQ}	MDC Frequency		25	MHz
t_{SETUP}	MDIO Setup to RX_CLK ring edge	5	-	ns
t_{HOLD}	RXD, RX_DV, RX_ER hold from RX_CLK ring edge	5	-	ns
T_{skew}	Data to Clock output Skew	0	30	ns

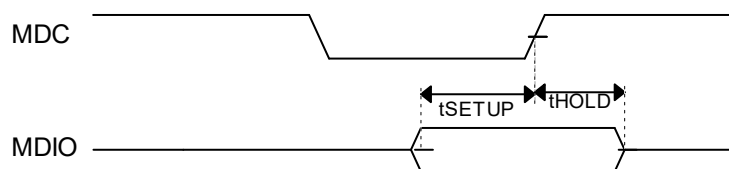


Figure 4-29 MDIO transmit signal timing diagram

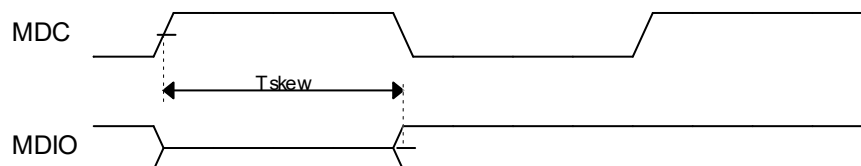


Figure 4-30 MDIO receive signal timing diagram

4.7.12 VIN timing parameters

4.7.12.1 VIN signals map

Table 4-53 VIN signals map

Ball Name	Description	Direction
VIN_LLC	Clock for VIN Pixel data	I
VIN_VS	VIN vertical synchronization	I
VIN_HS	VIN horizontal synchronization	I
VIN_DE	VIN data valid	I
VIN_R0	VIN pixel data0	I
VIN_R1	VIN pixel data1	I
VIN_R2	VIN pixel data2	I
VIN_R3	VIN pixel data3	I
VIN_R4	VIN pixel data4	I
VIN_G0	VIN pixel data5	I
VIN_G1	VIN pixel data6	I
VIN_G2	VIN pixel data7	I
VIN_G3	VIN pixel data8	I
VIN_G4	VIN pixel data9	I
VIN_G5	VIN pixel data10	I
VIN_B0	VIN pixel data11	I
VIN_B1	VIN pixel data12	I
VIN_B2	VIN pixel data13	I
VIN_B3	VIN pixel data14	I
VIN_B4	VIN pixel data15	I

4.7.12.2 VIN timing parameters

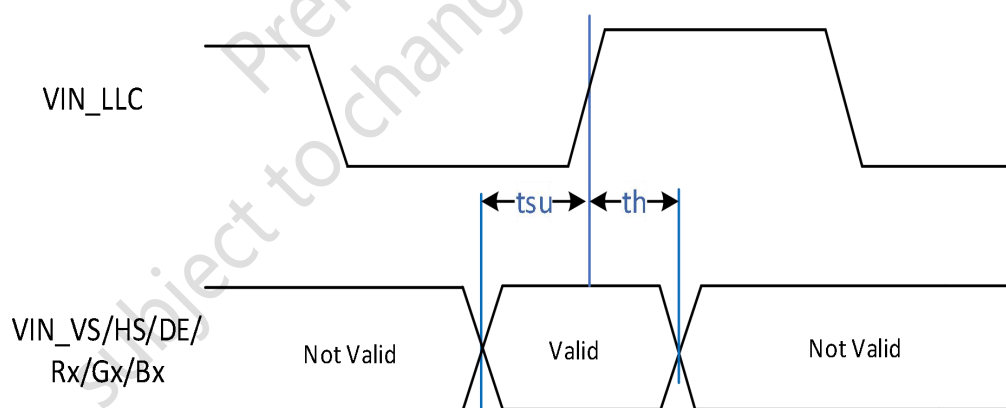


Figure 4-31 VIN timing diagram

Table 4-54 VIN timing parameters

Symbol	Parameter Description	MIN	Typical	Max	Unit
VIN_LLC	Input clock frequency	13.5		165	MHz
tsu	Input Data Setup time	2.11			ns
th	Input Data Hold time	0.21			ns

Note:

- If select ADV7611 as the HDMI receiver, the delay between clock input and data input must not exceed 200ps.

4.7.13 VOUT timing parameters

4.7.13.1 VOUT signals map

Table 4-55 VOUT signals map

Ball Name	Description	Direction
VOUT_R0	Display output data	O
VOUT_R1	Display output data	O
VOUT_R2	Display output data	O
VOUT_R3	Display output data	O
VOUT_R4	Display output data	O
VOUT_R5	Display output data	O
VOUT_R6	Display output data	O
VOUT_R7	Display output data	O
VOUT_G0	Display output data	O
VOUT_G1	Display output data	O
VOUT_G2	Display output data	O
VOUT_G3	Display output data	O
VOUT_G4	Display output data	O
VOUT_G5	Display output data	O
VOUT_G6	Display output data	O
VOUT_G7	Display output data	O
VOUT_B0	Display output data	O
VOUT_B1	Display output data	O
VOUT_B2	Display output data	O
VOUT_B3	Display output data	O
VOUT_B4	Display output data	O
VOUT_B5	Display output data	O
VOUT_B6	Display output data	O
VOUT_B7	Display output data	O
VOUT_HS	Display output horizontal sync	O
VOUT_VS	Display output vertical sync	O
VOUT_DE	Display output horizontal valid	O
VOUT_PCLK	Display output clock	O

4.7.13.2 VOUT timing parameters

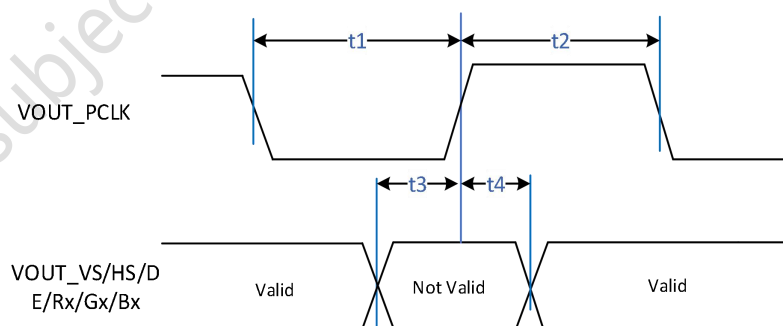


Figure 4-32 VOUT timing diagram

Table 4-56 VOUT timing parameters

Parameter	Symbol	Description	Min	Typ	Max	Unit
VOUT_PCLK Frequency Range			27		165	MHz

VOUT_PCLK Duty Cycle	t1:t2		45:55		55:45	
Data Output Transition Time	t3	End of valid data to negative clock edge			1.2	ns
	t4	Negative clock edge to start of valid data			0.8	ns

Note:

- If select ADV7513 as the HDMI Transmitter, the delay between clock input and data input must not exceed 200ps.

4.7.14 USB3 PHY parameters

The USB 3.0 femtoPHY supports the USB 3.0 SuperSpeed (5 Gbps) protocol and data rate and is backwardcompatible with USB 2.0 high-speed (480 Mbps), full-speed (12 Mbps), and low-speed (1.5 Mbps) protocols and data rates.

The USB 3.0 femtoPHY is designed to handle a wide range of input clock frequencies to support both host and device applications.

The following table summarizes the requirements of the reference clock provided to the USB 3.0 femtoPHY through the ref_pad_clk_(p,m) and ref_alt_clk_(p,m) input pins to support SuperSpeed and high-speed operations.

Table 4-57 Reference Clock Requirements

Symbol	Parameter	Min.	Typ.	Max.	Unit	Conditions
FREF_CLK	Reference clock frequency	19.2	-	200	MHz	-
FREF_OFFSET	Reference clock frequency offset	-300	-	300	ppm	-
RMSJREF_CLK	Reference clock random jitter (RMS)	-	-	3	ps	1.5 MHz to Nyquist frequency. For example, for 100-MHz reference clock, the Nyquist frequency is 50 MHz.
REF_CLK_SKEW	Reference clock skew	-	-	200	ps	-
DJREF_CLK	Reference clock cycle-to-cycle jitter	-	-	150	ps	DJ over all frequency
DCREF_CLK	Duty cycle	40	-	60	%	-
VCMREF_CLK	Common mode input level	(vp - 10%) / 2	-	(vp + 10%) / 2	V	Differential inputsa
VDREF_CLK	Differential input swing	2 x (vp - 10%)	-	2 x (vp + 10%)	Vpp	Differential inputsa
VOLREF_CLK	Single-ended input logic: Low	0	-	0.3	V	If single-ended input is used
VOHREF_CLK	Single-ended input logic: High	vp-0.3	-	vp	V	If single-ended input is used
SWREF_CLK	Input edge rate	0.6	-	4	V/ns	-
R_resref	Required external	-	200	-	ohm	± 1% accuracy

	reference resistance					
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4.7.15 USB2 PHY parameters

USB 2.0 PHY parameters are compatible with USB 3.0 PHY. Please refer to Section 4.7.14 USB 3.0 PHY parameters for more detailed information.

4.7.16 PCIE PHY parameters

4.7.16.1 Reference Clock: Jitter Requirements

The PHY supports a differential reference clock source. The source may be driven through either external pads or internal pins. The reference clock must meet specific requirements for signal swing and jitter.

The table below lists reference clock jitter requirements for all supported PCIe clock architectures.

Table 4-58 Reference Clock Jitter Requirements for PCIe 1.1 and 2.1

Application	Reference Clock Jitter Requirements
PCIe 1.1 - 2.5 Gbps	Refer to the PCI Express 3.0 Card Electromechanical specification.
PCIe 2.1 - 5.0 Gbps	Refer to the PCI Express 3.0 base specification.

Table 4-59 Reference Clock Jitter Requirements for PCIe 3.1

Symbol	Parameter	Max	Units	Conditions
RjREFCLK PCIe 3.1 - 8Gbps	Reference clock random jitter	3	ps, rms	Integrated Rj from 12kHz to 20MHz
		2.2		Integrated Rj from 2MHz to 20Mhz
DjREFCLK PCIe 3.1 - 8Gbps	Reference clock deterministic jitter	5.7	ps, pp	200KHz to 100MHz
		2.8		1MHz to 20MHz

4.7.16.2 Reference Clock: Electrical Characteristics

Table 4-60 Reference Clock Electrical Characteristics

Symbol	Parameter	Min	Max	Units	Conditions
FREF_OFFSET	Reference clock frequency offse	-150	150	ppm	
DCREF_CLK	Duty cycle	40	60	%	
	Maximum input voltage level		VP	V	Applies to ref_alt_clk_p/m only(CMOS level)
	Minimum input voltage level	0		V	
	Maximum input voltage level		0.88 ^a	V	Applies to ref_pad_clk_p / ref_pad_clk_m only
	Minimum input voltage level	-0.15		V	
	Absolute maximum voltage at pin including overshoot		1.15	V	
	Absolute minimum voltage at pin including overshoot	-0.3		V	
	Maximum single-ended input swing		0.88	V	
SWREF_CLK	Input edge rate	0.4	2.0	V/ns	Applies to ref_pad_clk_p/m only

REF_CLK_SKEW	Reference clock skew ($\pm$)	200	ps
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a. Allowed overshoot time: 10% of period.

4.7.16.3 Reference Clock External Pad Input Parasitics

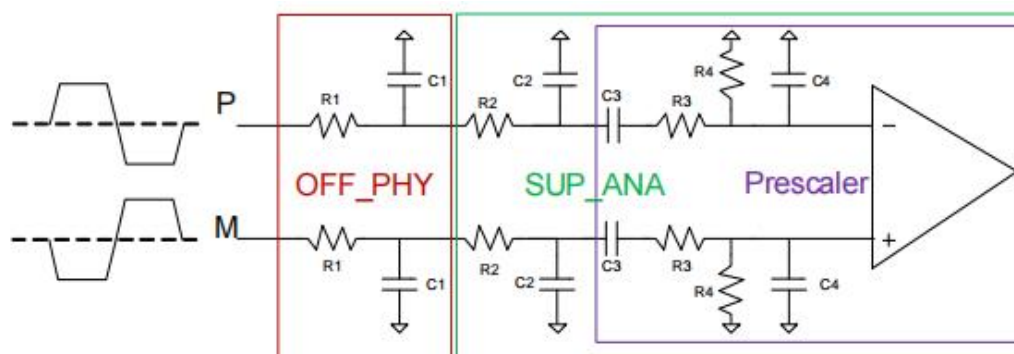


Figure 4-33 Reference Clock External Pad Input Parasitics

Table 4-61 Reference Clock External Pad Input Parasitics

Location	Parasitics Device	Value	Unit
OFF PHY	R1	≤ 50	Ω
OFF PHY	C1	≤ 1	pF

4.7.16.4 Reference Resistor Requirements

Reference resistor requirements are outlined in Table 4-63.

Table 4-62 Reference Resistor Requirements

Item	Value	Design Guidelines
External Reference Resistor Resistance	200 ohm precision resistor (+/-1%)	None
External Reference Resistor Maximum Current	2mA	None
External Reference Resistor parasitic capacitance C_{load}	< 10 pF	It is important to meet the total capacitance at low frequency (in other words, the total capacitance of pad, package trace, and PCB).
External Reference Resistor parasitic resistance through package and board R_{route}	1% of 200 ohm	Take care to minimize the resistance of the connection through the package to the external reference resistor as this will affect the accuracy of the termination tuning. This spec is for total resistance through the package and board to the board external resistor. The ground used on the board to terminate the external resistor should be as close as possible to the PMA ground voltage. Noise adjacent signals, for example clocks, should be spaced away from resref or shielded with ground.

4.7.16.5 External Components

The PHY requires the following external components:

- AC coupling capacitors as specified by the standard, for reference, see Table 3-12
- Supply bypass capacitors selected to minimize supply noise. Supply bypass capacitor are recommended to minimize power supply noise. Noise analysis of the power delivery network is required to determine the actual values. Depending on their size, each capacitor will have a different equivalent series resistance (ESR) and equivalent series inductance (ESL) that will determine the given capacitor's effectiveness over a frequency range. In general, several low-value capacitors should be placed as close as possible to the package pins. Larger-value capacitors can be placed farther away. Table 3-13 provides an example list of typical capacitor values.
- Precision 200 ($\pm 1\%$) resistor used for biasing and termination impedance calibration; for more information, see 4.7.16.4 and Table 4-66.

Table 4-63 AC Coupling Capacitors - Consumer 10G AP PCIe 3.0 PHY Protocol

Protocol	Min	Typ	Max	Units
PCIe	180	--	265	nF

Table 4-64 Example Supply Bypass Capacitors

Component	Value ^a
Power Supply Bypass Capacitors	0.01 uF, 0.1 uF, 4.7 uF, 10.0 uF

a. Smaller-value capacitors must be placed between the ferrite bead and the package.

Table 4-65 External Component Requirements

Component	Value
resref	200 ohm (+/- 1%)

4.7.17 MIPI DPHY parameters

The following table describes the High Speed Clock Timing.

Table 4-66 High Speed Clock Timings

Symbol	Parameter	Condition	Mini.	Typ.	Max.	Unit	Notes
-	Maximum Serial Data rate (forward direction)	On DATAP/N outputs. $80\Omega \leq RL \leq 125\Omega$	80		2500	Mbps	
FDDRCLK	DDR CLK frequency	On CLKP/N outputs	40		1250	MHz	
TDDRCLK	DDR CLK period	$80\Omega \leq RL \leq 125\Omega$	0.8		25	ns	
UIINST	UI instantaneous		0.5		12.5	ns	This value corresponds to a minimum Mbps data rate.
ΔUI	UI variation		-10%		10%	UI	When $UI \geq 1ns$, within a single burst.
			-5%		5%	UI	When $0.667ns < UI < 1ns$, within a single burst.
tCDC	DDR CLK	tCDC =		50		%	

	duty cycle	tCPH / TDDRCLK					
tCPH	DDR CLK high time			1		UI	
tCPL	DDR CLK low time			1		UI	
-	DDR CLK / DATA Jitter			75		ps pk-pk	When UI < 1ns, within a single burst.
TSKEW[TX]	Data to Clock Skew (TX)		-0.15		0.15	UI	For PHY operating at or below 1.5Gbps
TSETUP[RX]	Data to Clock Setup Time (RX)		0.15			UI	For PHY operating at or below 1Gbps
			0.2			UI	For PHY operating above 1Gbps and below or at 1.5Gbps

5 Boot configuration

5.1 Boot mode configuration pins

The boot process of BST A1000L decides the ultimate security and safety apparatus of the ECU. The SOC supports three main boot modes summarized in the following tables. The boot modes are selected via boot mode pins at power-on. Boot_mode_sel determines the main boot mode. Boot mode does not change until the device is power cycled.

Table 5-1 Boot Mode Configuration

Boot Mode	Security Sub-block	Safety Sub-block	App proc & SOC
Boot from Nor-flash with Hardware Authorize ¹ (3'b000)	Boots from on-chip ROM with Authorize and AB Zone Check	Boots by security	Boots by security
Boot from Nor-flash (3'b100)	Boots Nor-flash	Boots by security	Boots by security
Boot from Emmc (3'b101)	Boots from Emmc	Boots by security	Boots by security
Safe Boot ² (3'b001)	Bypassed	Boots from nor-flash	Boots by security
App_dev Boot ³ (3'bx1x)	Bypassed	Bypassed	Boots from nor-flash

Note:

1. Boot from Norflash with Hardware Authorize is the boot mode with hardware verification and is the most secure boot mode
2. Safe boot skips the verification process and will boot from safety subsystem
3. App_dev boot is the boot mode of A55 from flash, and without any security measures

5.2 Boot device interface allocation

The following table lists the interfaces that can be used by the boot process in accordance with the specific Boot mode configuration.

Table 5-2 Interface allocation during boot

Interface	IP Instance	Allocated Pads During Boot	Comment
OSPI Flash	OSPI0/OSPI1	OSPI0_CS0, OSPI0_SCLK, OSPI0_IO0~7/ OSPI1_CS0, OSPI1_SCLK, OSPI1_IO0~7	A1000L will boot from OSPI1 if the OSPI0 boot failed

5.3 Boot strap pin configuration

The following table lists the strap pin that can be used by the boot process to configure system mode.

Table 5-3 Boot strap pin configuration

Ball Name	Strap Pin Name	Description	Default Value	Comment
GPIO_24	r5_lock_step	Safety R5 work on lock-step mode 1: enable	1	

		0: disable		
I2S0_CK	sys_parity_en	System parity check enable 1: enable 0: disable	1	
SPI1_SCLK	sys_ecc_en	System ecc check enable 1: enable 0: disable	1	
GPIO_107	ist_post_bypass	On-chip dma-based bist 1: bypass 0: no bypass	0	
GPIO_26	ist_post_auto_mode	1: auto mode 0: manual mode	1	Manual mode is used for debug only
GPIO_29	boot_mode_sel[2]	Boot mode select 3'b000: sec boot from nor-flash with hardware authorize 3'b001: safety boot 3'bx1x: app dev boot 3'b100: sec boot from nor-flash 3'b101: sec boot from EMMC	100	
GPIO_28	boot_mode_sel[1]			
GPIO_27	boot_mode_sel[0]			
GPIO_25	usb_host_dev	1: USB Host 0: USB Device	0	
I2S0_MCK	pcie_bifurc_en	1: enable PCIE bifurc 0: disable PCIE bifurc	1	For detailed guidance, please ref to "Table 5-4 Strap PINs for PCIe"
GPIO_108	pcie_x4_is_rc	1: PCIE RC 0: PCIE EP	1	

Table 5-4 Strap PINs for PCIe

Strap PINs for PCIe		PCIe Controller 0				PCIe Controller 1			
I2S0_MCK (bifurcation_en)	GPIO_108 (x4_is_rc)	Enabled	Max Link Width	PHY Lanes Assigned	Mode	Enabled	Max Link Width	PHY Lanes Assigned	Mode
0	0	Yes	4	lane 0, 1, 2, 3	EP	No	-	None	EP
0	1	Yes	4	lane 0, 1, 2, 3	RC	No	-	None	EP
1	0	Yes	2	lane 0, 1	EP	Yes	2	lane 2, 3	EP
1	1	Yes	2	lane 0, 1	RC	Yes	2	lane 2, 3	EP

6 PCB Guideline or Requirement

6.1 PCB Guideline

The PCB Guideline or Requirement please refer to *BST A1000 Hardware Design Guidelines.pdf*.

6.2 Unused pin

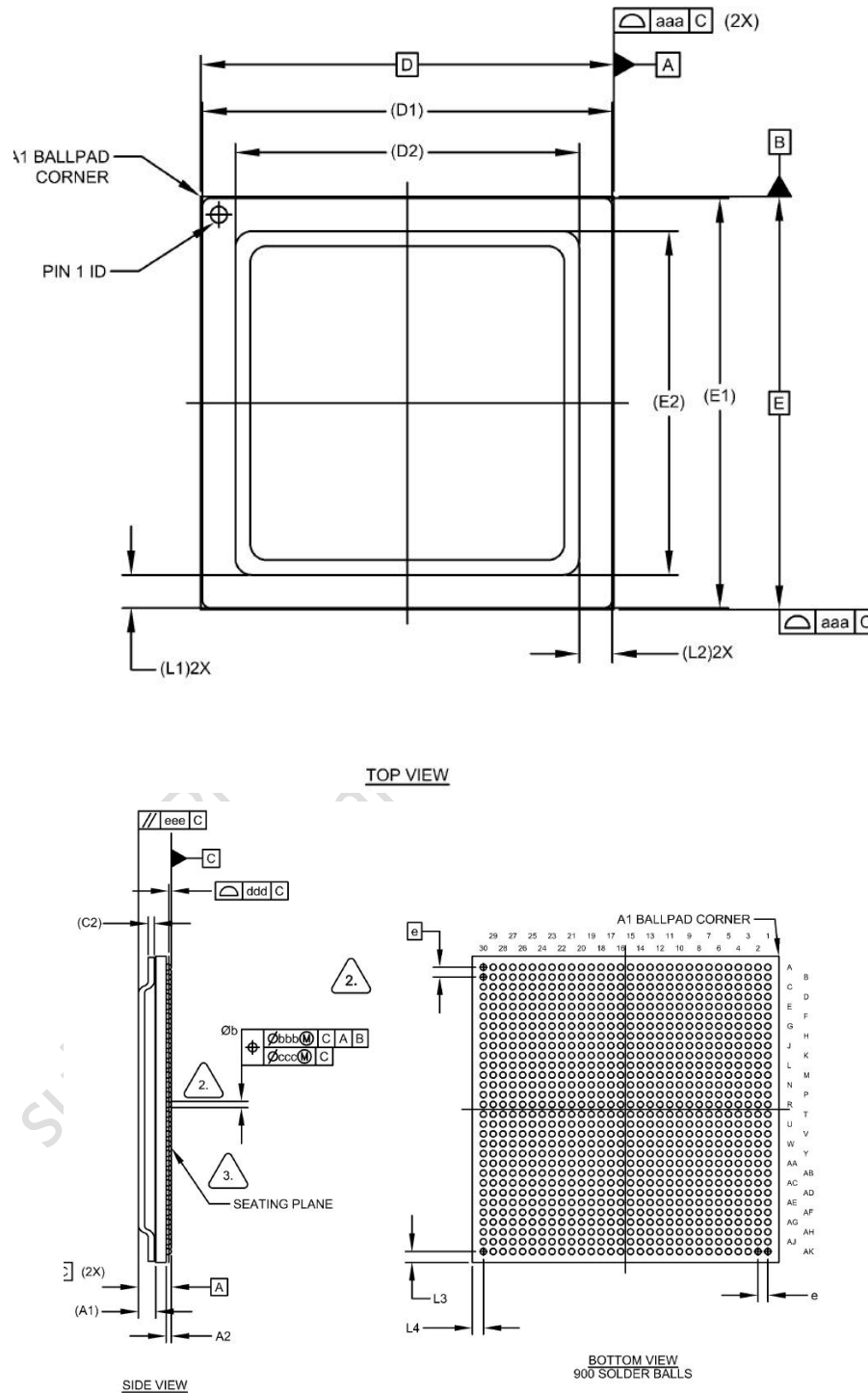
Please refer to the latest *A1000_Unsed_Pin_Configuration.xlsx* to deal with the unused pin.

Preliminary Version
subject to change without notification

7 Package information and contact assignments

7.1 Package outline Dimension

Figure 7-1 shows the outline dimension of the 25×25 mm, 0.8 mm pitch FCBGA package.



ITEM	DIMENSION IN MM		
	MIN	NOM	MAX
D	24.800	25.000	25.200
E	24.800	25.000	25.200
A	2.442	2.670	2.899
A1	---	1.410	---
A2	0.300	0.400	0.500
D1	---	24.800	---
D2	---	20.800	---
E1	---	24.800	---
E2	---	20.800	---
L1/L2	---	2.000	---
L3/L4	---	2.000	---
b	0.460	0.510	0.560
e	---	0.800	---
aaa	0.200		
bbb	0.200		
ccc	0.080		
ddd	0.200		
eee	0.350		

Figure 7-1 Package outline dimension

7.2 Supplies contact assignments and functional contact assignments

Table 7-1 shows supplies contact assignments for A1000L.

Table 7-1 A1000L supplies contact assignments

Power Rail Name (BALL NAME)	Remark
DDR0_VAA	DDR0 PLL supply voltage connected to independent 1.8V supply
DDR0_VDDQ	DDR0 VDDQ power supply
DDR1_VAA	DDR1 PLL supply voltage connected to independent 1.8V supply
DDR1_VDDQ	DDR1 VDDQ power supply
MIPI_VP	MIPI Analog 0.8V Supply
MIPI_VPH	MIPI Analog 1.8V Supply
OTP_VDD2	OTP supply
PCIE0_VP	PCIE PHY0 Analog supply
PCIE0_VPH	PCIE PHY0 High Voltage IO Supply
PCIE1_VP	PCIE PHY1 Analog supply
PCIE1_VPH	PCIE PHY1 High Voltage IO Supply
PLL_AON_VDDHV	PLL to safety VDDHV supply
PLL_DDR0_VDDHV	PLL to DDR0 VDDHV supply
PLL_DDR1_VDDHV	PLL to DDR1 VDDHV supply
PLL_PCIE_VDD	PLL to PCIE VDD supply
PLL_PCIE_VDDHV	PLL to PCIE VDDHV supply
PLL_TOP_VDD	PLL to TOP VDD supply
PLL_TOP_VDDHV	PLL to TOP VDDHV supply

PVT_VDDA_1	PVT1 Sensor AVDD supply
PVT_VDDA_2	PVT2 Sensor AVDD supply
PVT_VDDA_3	PVT3 Sensor AVDD supply
PVT_VDDA_4	PVT4 Sensor AVDD supply
PVT_VDDA_5	PVT5 Sensor AVDD supply
PVT_VSSA_1	PVT1 Sensor AVDD Return path
PVT_VSSA_2	PVT2 Sensor AVDD Return path
PVT_VSSA_3	PVT3 Sensor AVDD Return path
PVT_VSSA_4	PVT4 Sensor AVDD Return path
PVT_VSSA_5	PVT5 Sensor AVDD Return path
USB20_DVDD	USB2 PHY analog and digital high-speed, low-voltage supply
USB20_VDD33	USB2 High-voltage supply for high-speed operation
USB20_VP	USB2 PHY analog and digital high-speed, low-voltage supply
USB20_VPTX	USB2 PHY transmit supply
USB30_DVDD	USB3 PHY analog and digital high-speed, low-voltage supply
USB30_VDD33	USB3 High-voltage supply for high-speed operation
USB30_VP	USB3 PHY analog and digital high-speed, low-voltage supply
USB30_VPTX	USB3 PHY transmit supply
VDD	SOC core supply
VDD_DDR0_MIPI	DDR0/MIPI core supply
VDD_DDR1	DDR1 core supply
VDD_GPU	GPU core supply
VDD_ISP_CV	ISP/CV core supply
VDD_NET	NET core supply
VDD_VSP	VSP core supply
VDD18_SDIO_00	Internal 1.8V supply output for SDIO IO0
VDD18_SDIO_01	Internal 1.8V supply output for SDIO IO0
VDD18_SDIO_02	Internal 1.8V supply output for SDIO IO0
VDD18_SDIO_10	Internal 1.8V supply output for SDIO IO1
VDD18_SDIO_11	Internal 1.8V supply output for SDIO IO1
VDD18_SDIO_12	Internal 1.8V supply output for SDIO IO1
VDDIO_AON	AON IO supply
VDDIO_SDIO_0	SDIO IO0 supply
VDDIO_SDIO_1	SDIO IO1 supply
VDDIO_SOC	SOC IO supply
VSS	core supply and IO supply return path

Note:

1. It is required to implement 0.5uF-1uF capacitor on VDDPST18 pin on PCB to stabilize the output voltage of LDO.

7.3 Pinmux and Ball map

The pinmux and ball map please refer to *A1000x-Bx IOMUX and Ballmap Configuration.xlsx*.